

SCK-KR-05101

Robotics Starter Kit Carrier Card

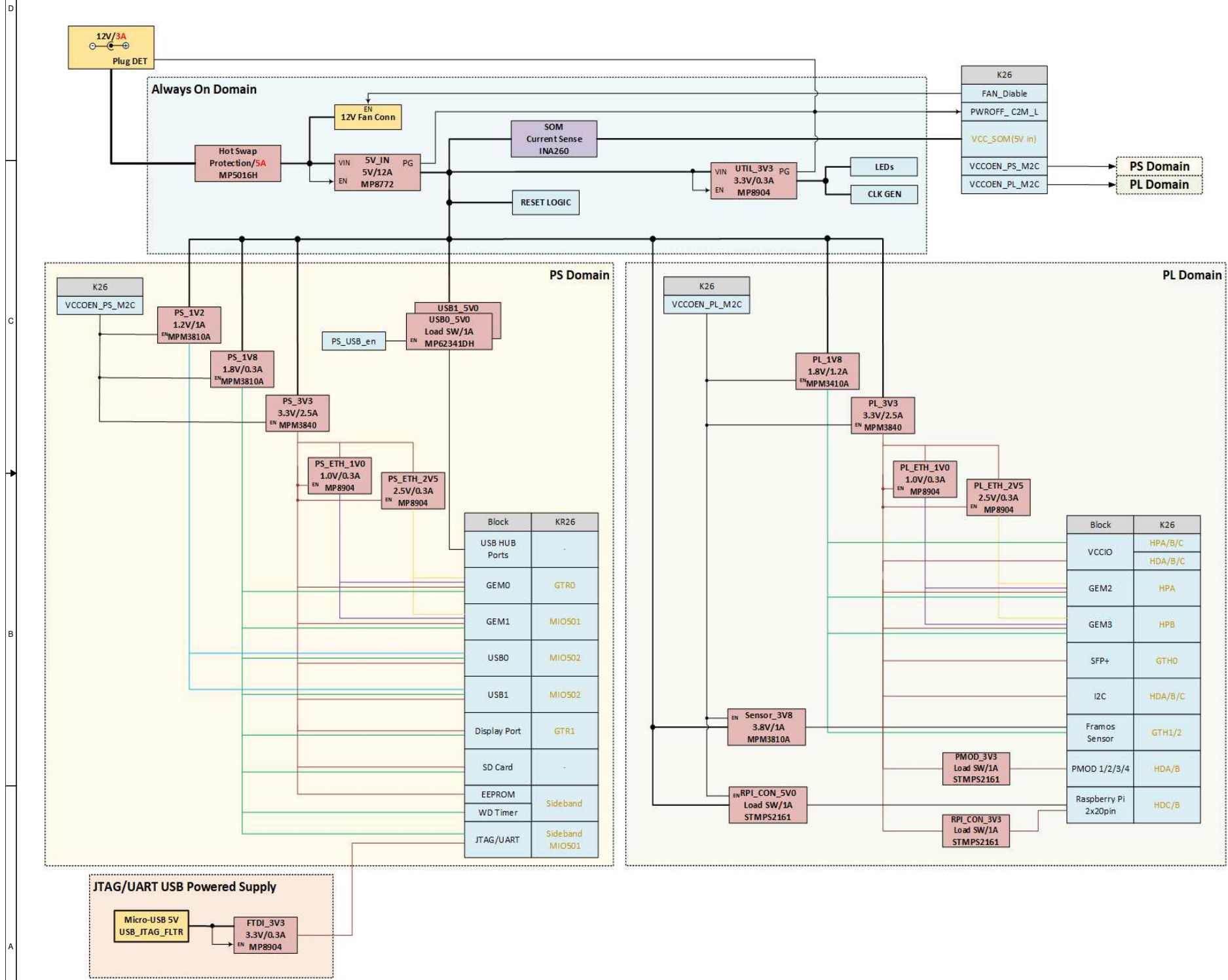
NOTICE: These schematics are made available for users who are working with this specific Xilinx board and should not be considered an optimal board design reference. This board was designed for evaluation purposes with components that are potentially more complex and costly than would be used in an end customer system. Users are encouraged to leverage the Xilinx design guidelines as the optimal starting point for new board designs.

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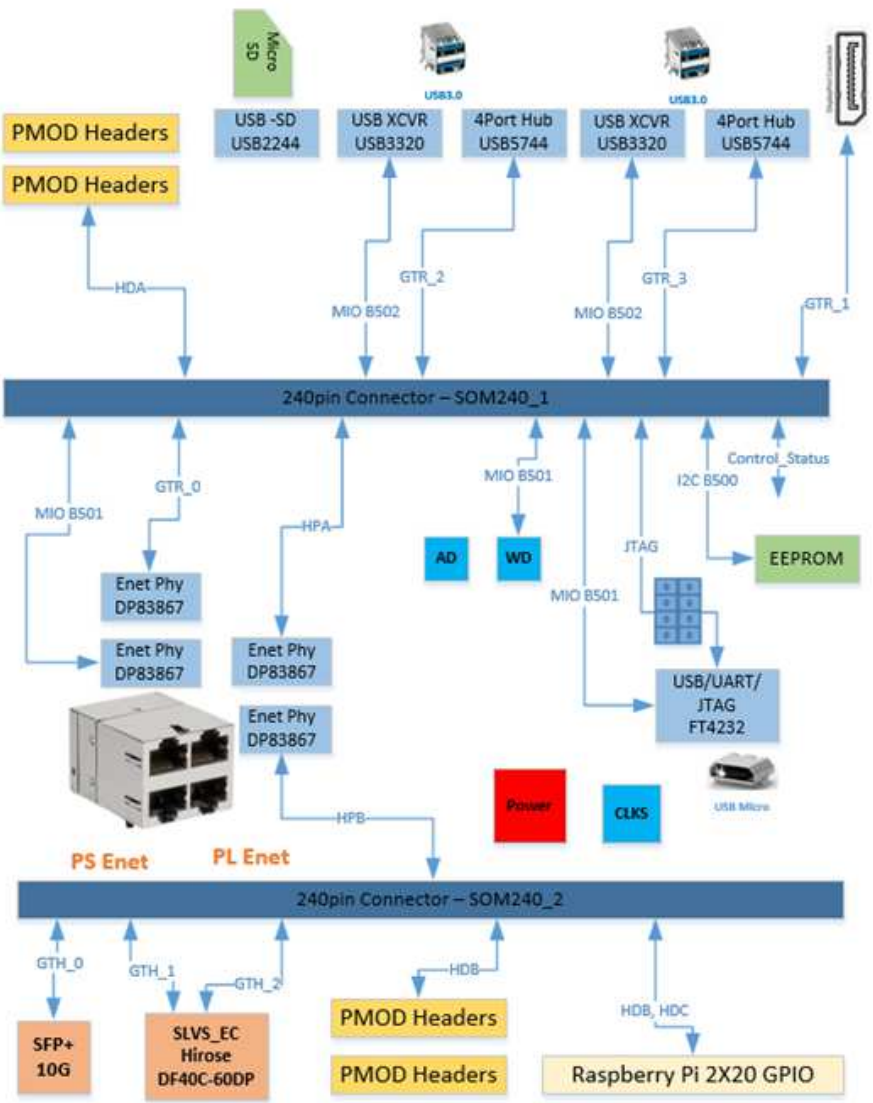
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 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 01 - Title Page			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
Engineer: RH/JW			Sheet 1 of 25

POWER BLOCK DIAGRAM

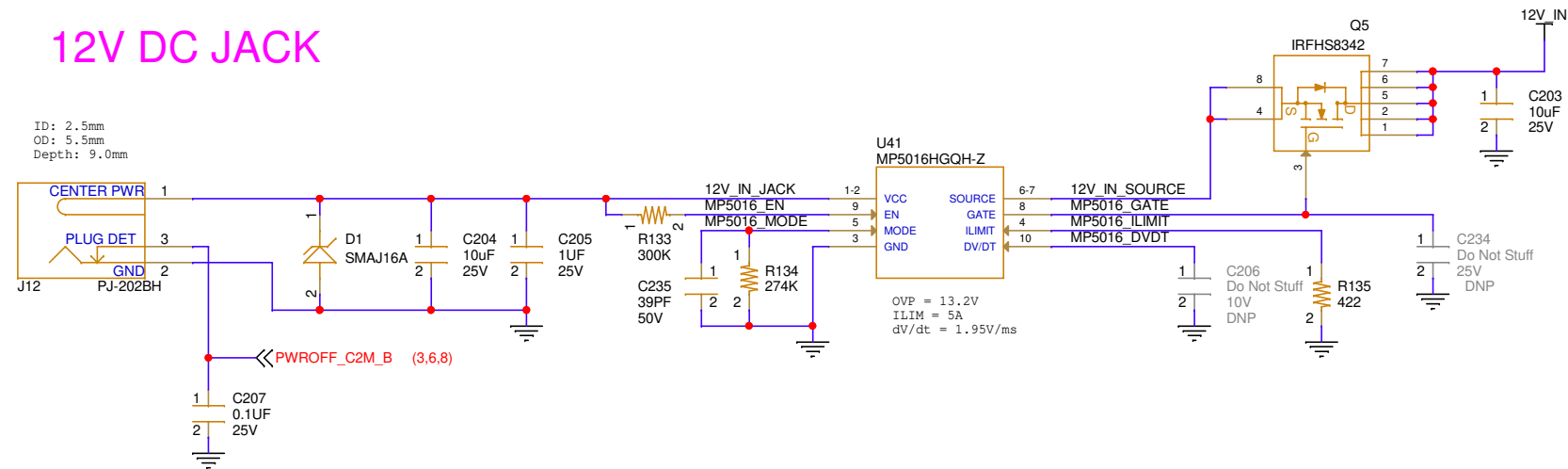


PERIPHERAL BLOCK DIAGRAM



 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 02 - Block Diagram			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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ID: 2.5mm
OD: 5.5mm
Depth: 9.0mm



12V IN

MMSD4148T1G

D2

1 2

Q18
NDS331N

6D0UFB4-7

C208
10µF
25V

L8
FERRITE-600

L9
FERRITE-600

12V_FAN_PWR

12V_FAN_GND

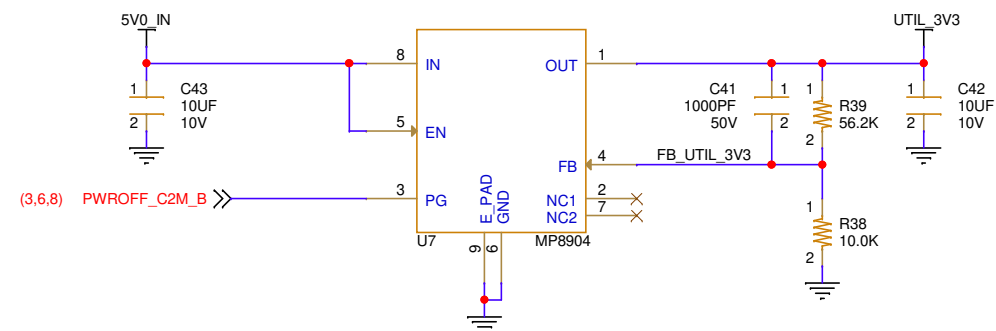
J13

C209
25V
DNP

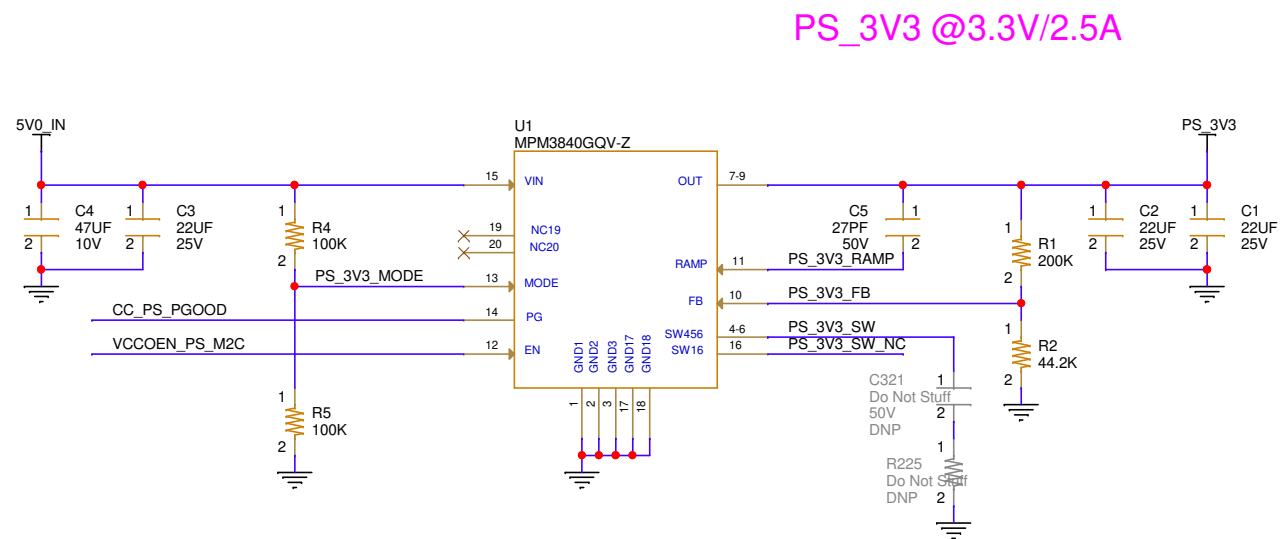
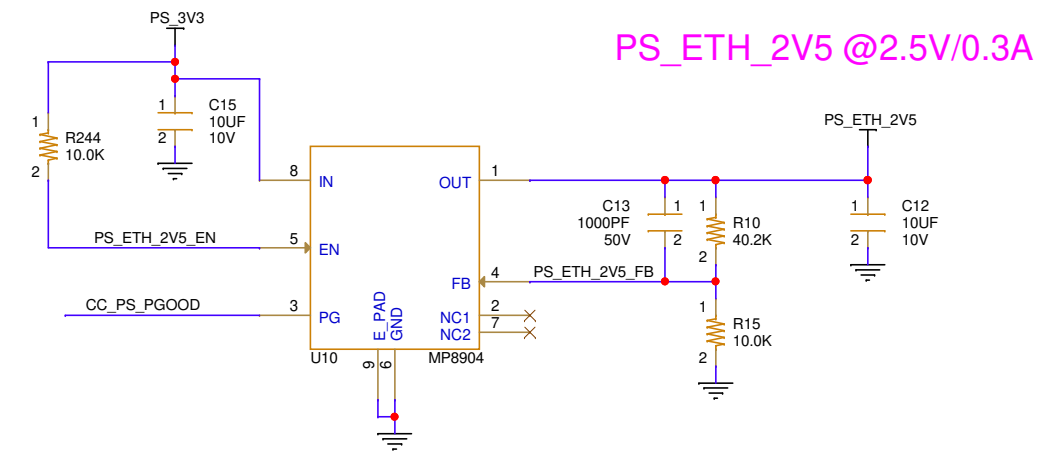
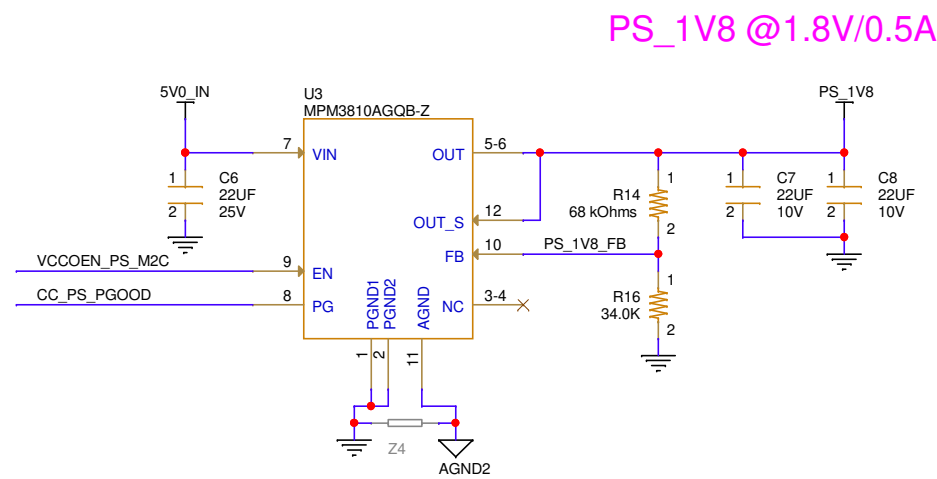
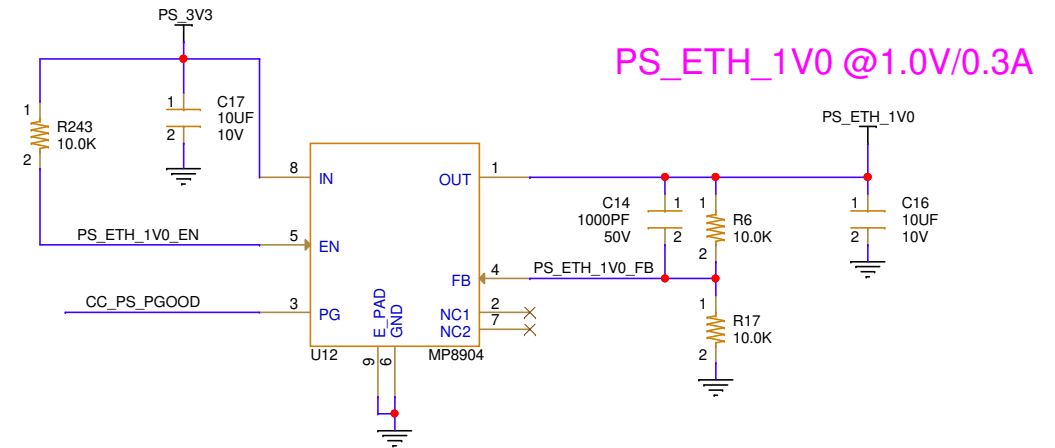
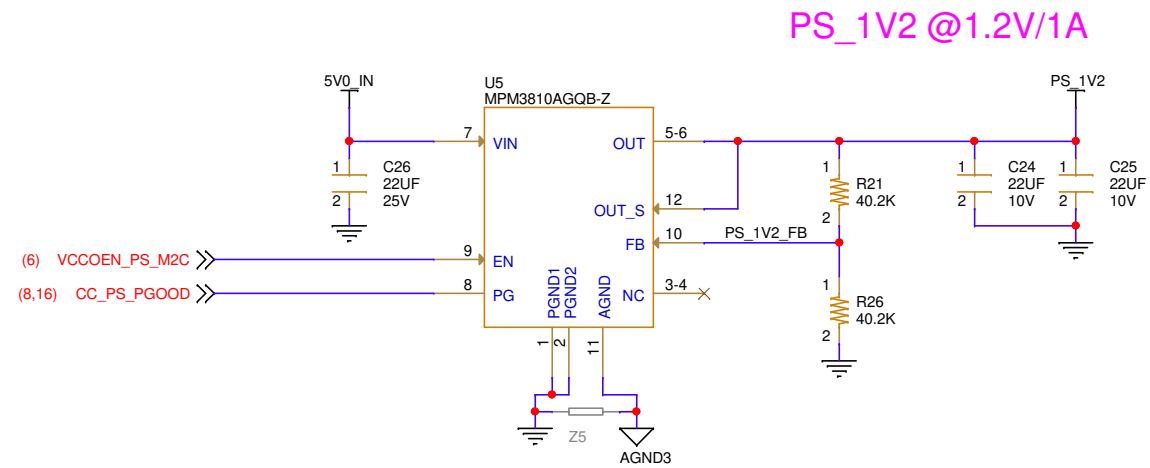
C210
25V
DNP

[illegible]

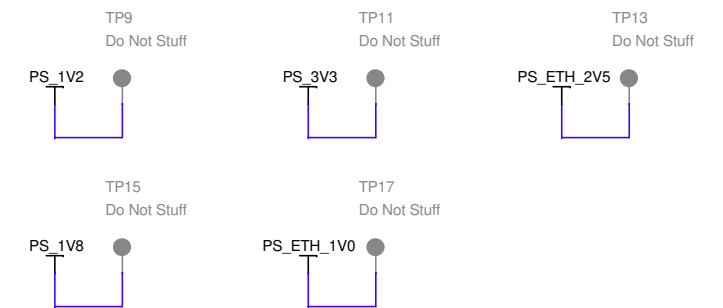
UTIL_3V3@3.3V/0.3A



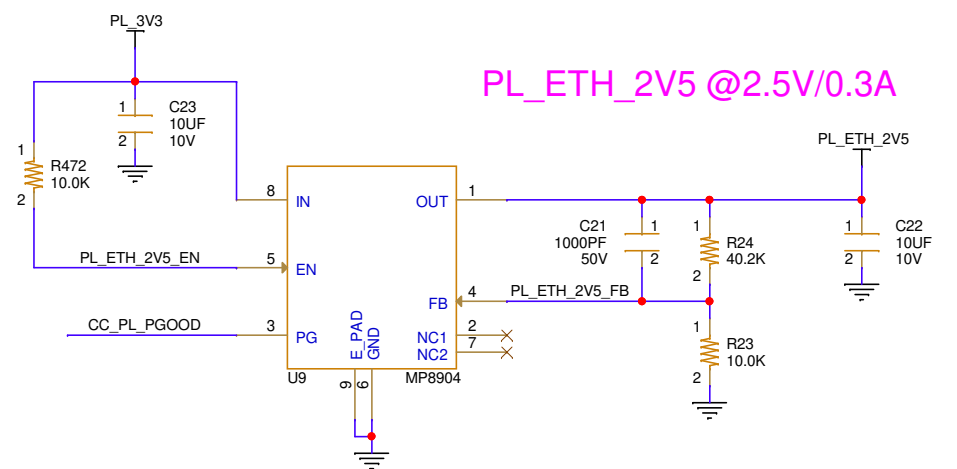
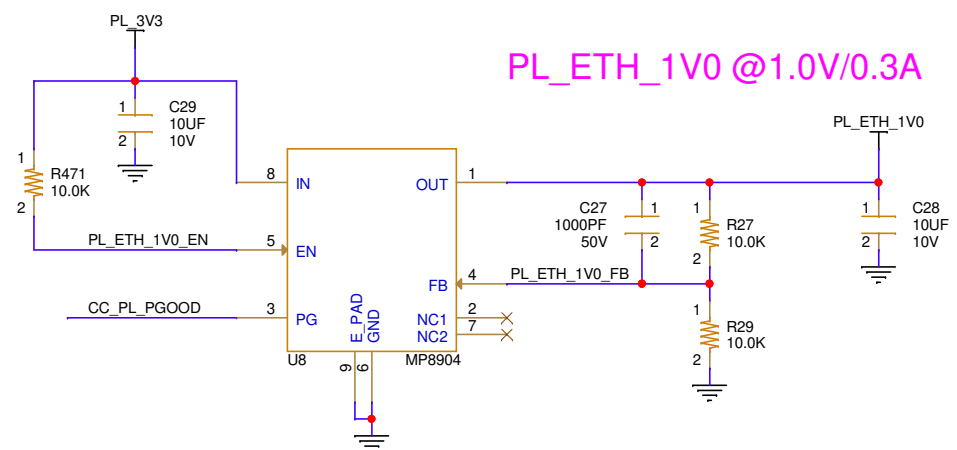
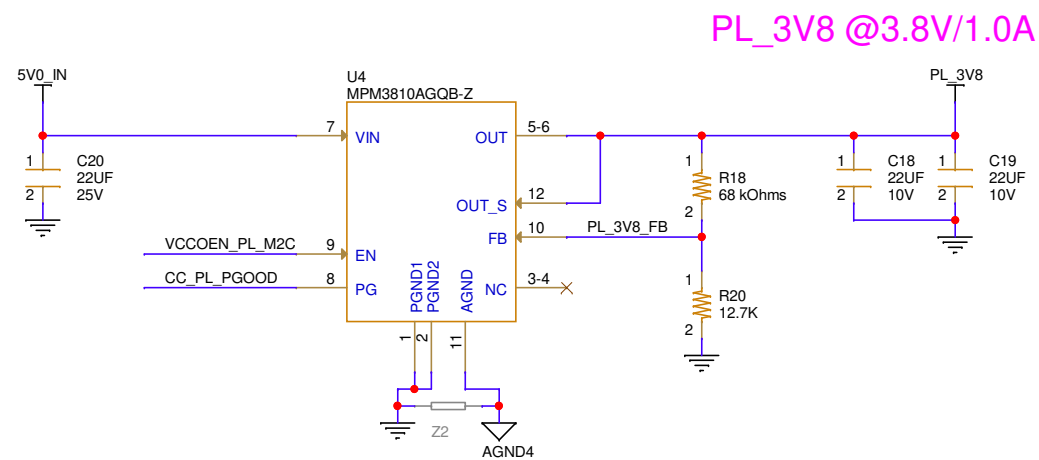
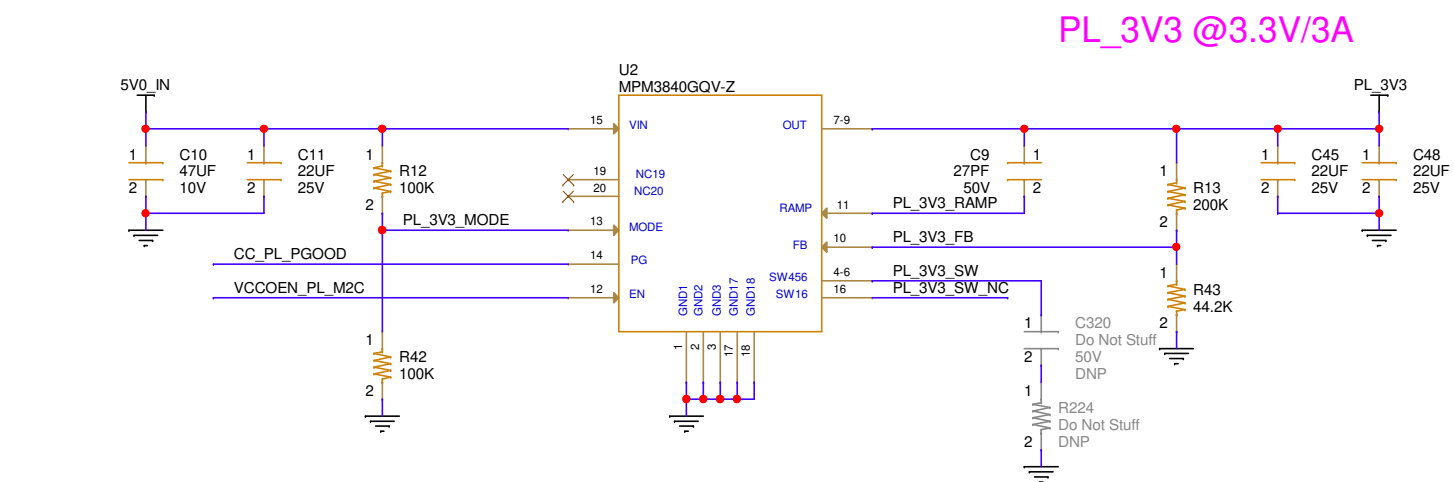
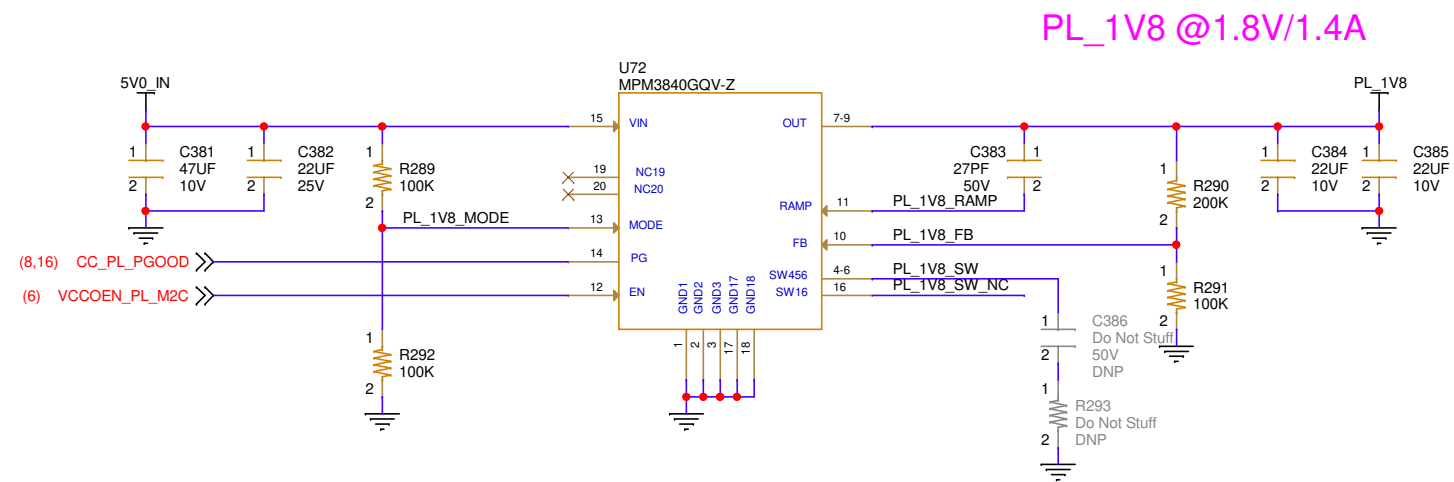
 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 03 - Always On Pwr, Fan Header			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
Engineer: RH/JW		Sheet 3 of 25	



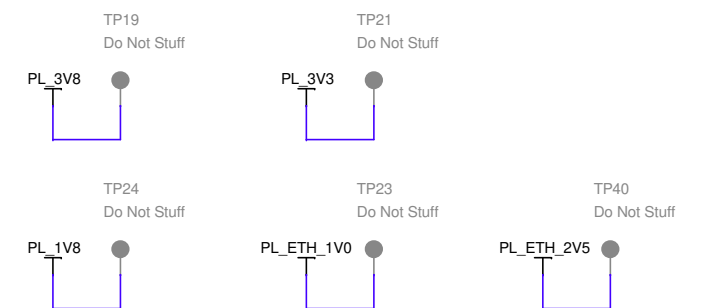
POWER TEST POINTS



 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT
Title: SCK-KR 04 - PS Power		
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01
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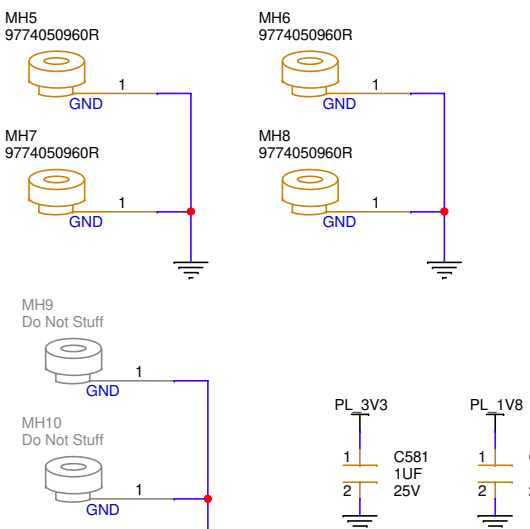
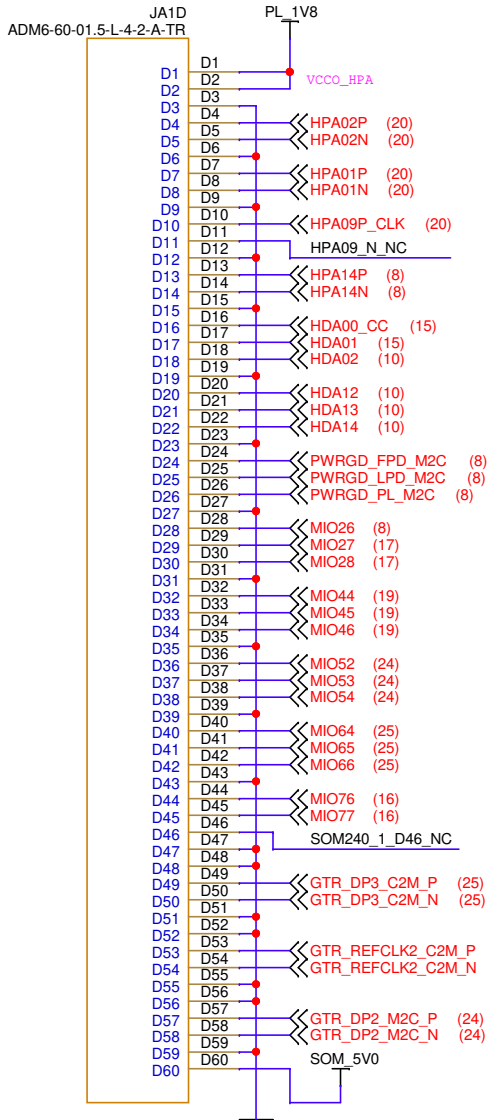
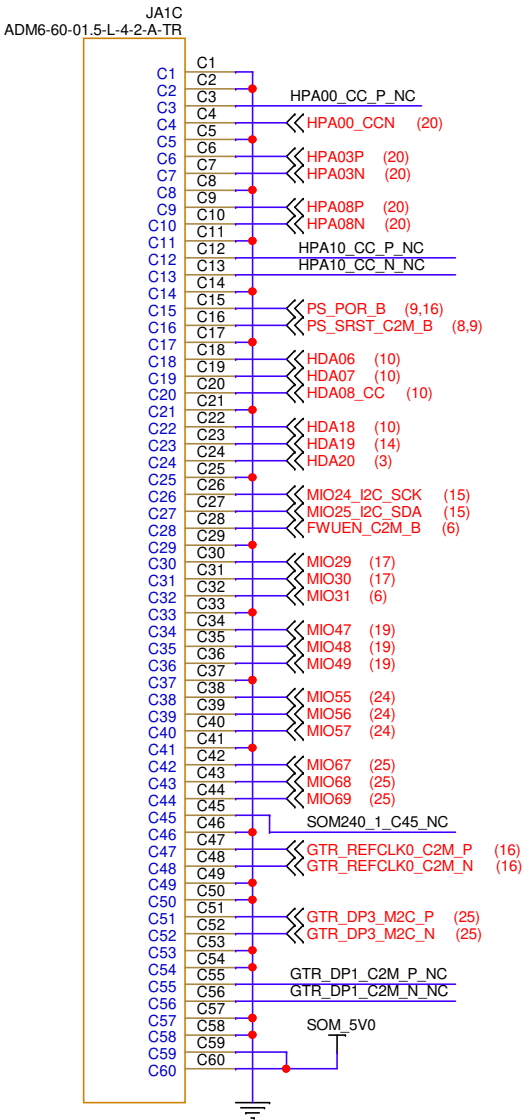
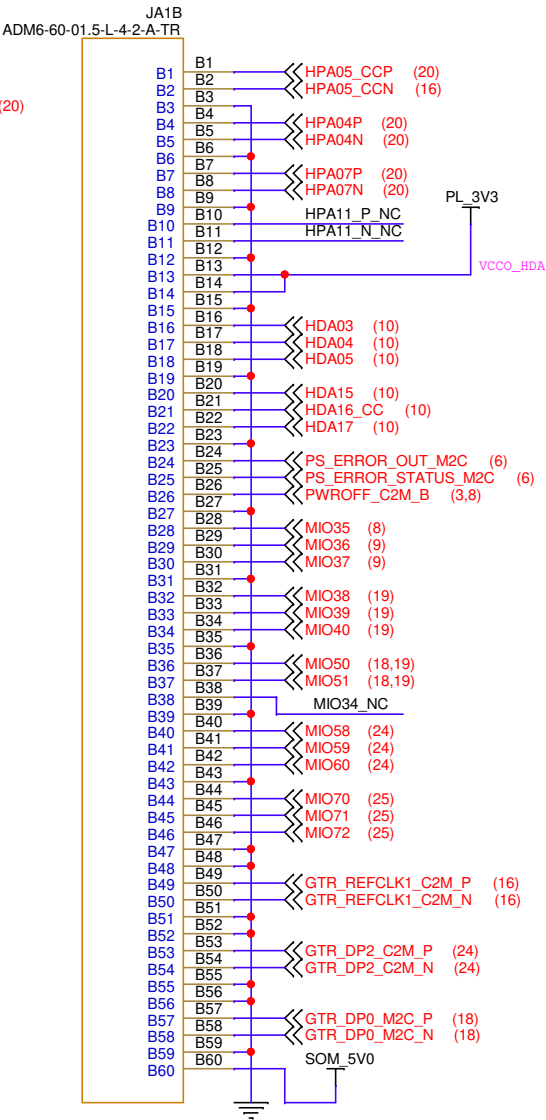
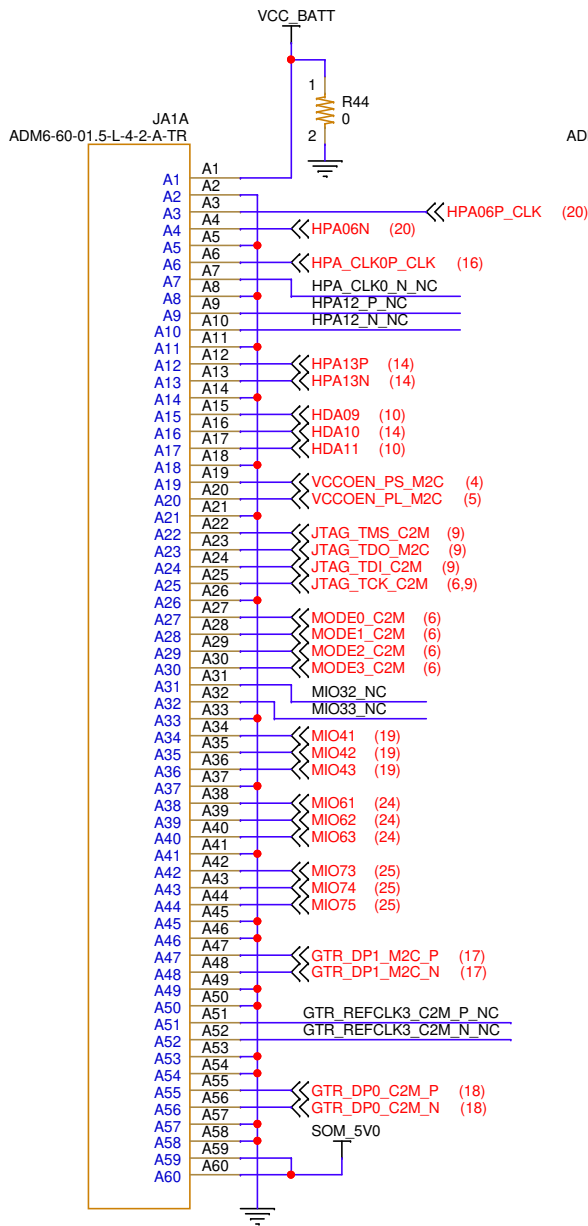
POWER TEST POINTS



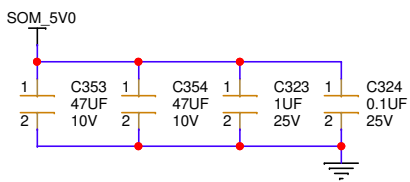
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Title: SCK-KR 05 - PL Power		
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01
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SOM240_1 CONNECTOR

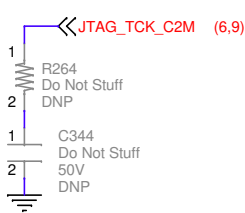
SOM STANDOFFS



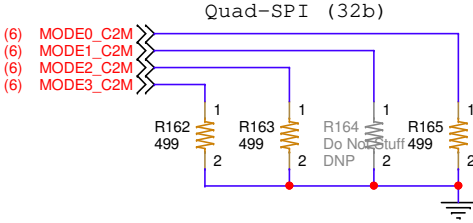
SOM_5V0 Connector Decoupling



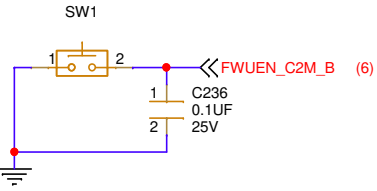
TCK TERM



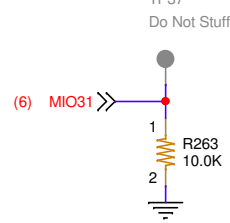
MODE PINS



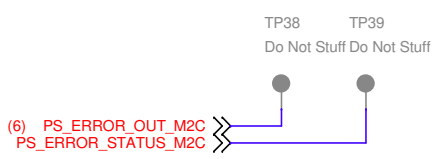
FWUEN BUTTON



SHUTDOWN



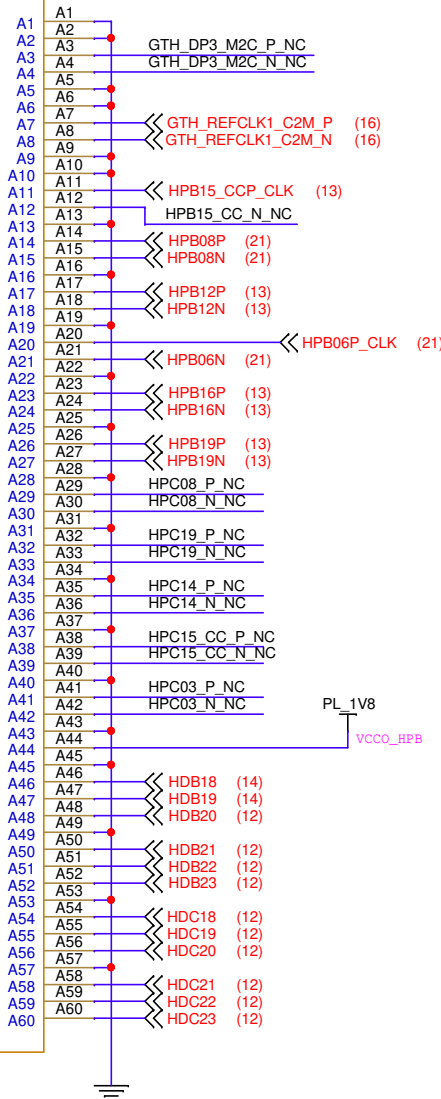
PS_ERROR_OUT/STATUS TEST POINTS



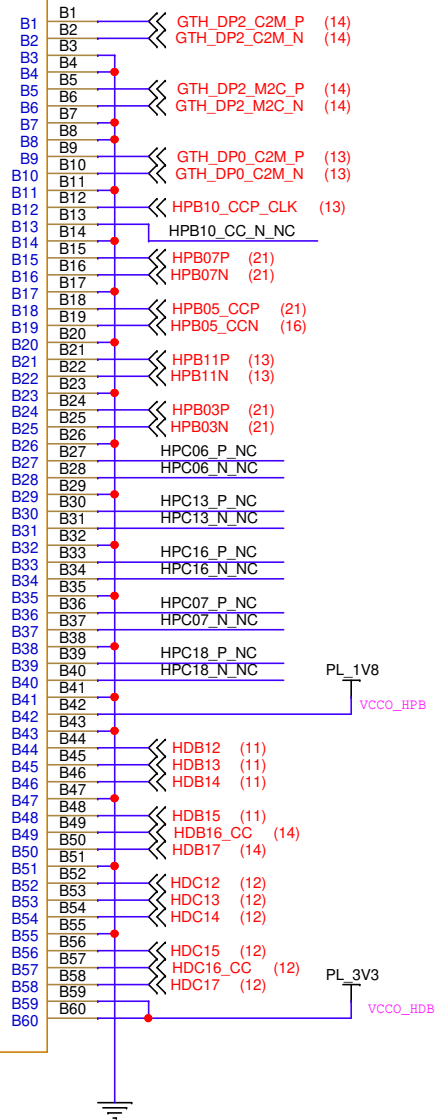
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Title: SCK-KR 06 - SOM240_1 Connector			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
Engineer: RH/JW		Sheet 6 of 25	

SOM240_2 CONNECTOR

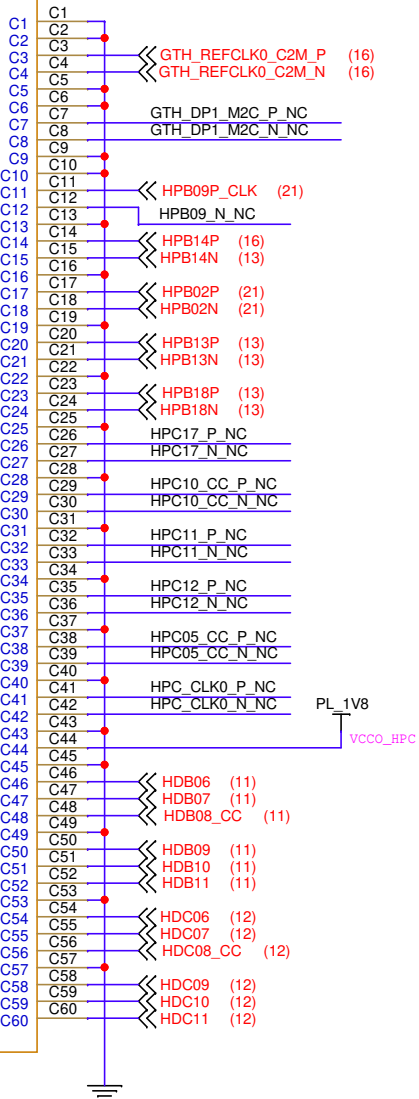
JA2A
ADM6-60-01.5-L-4-2-A-TR



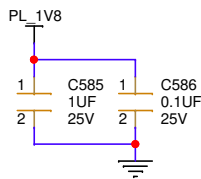
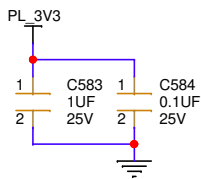
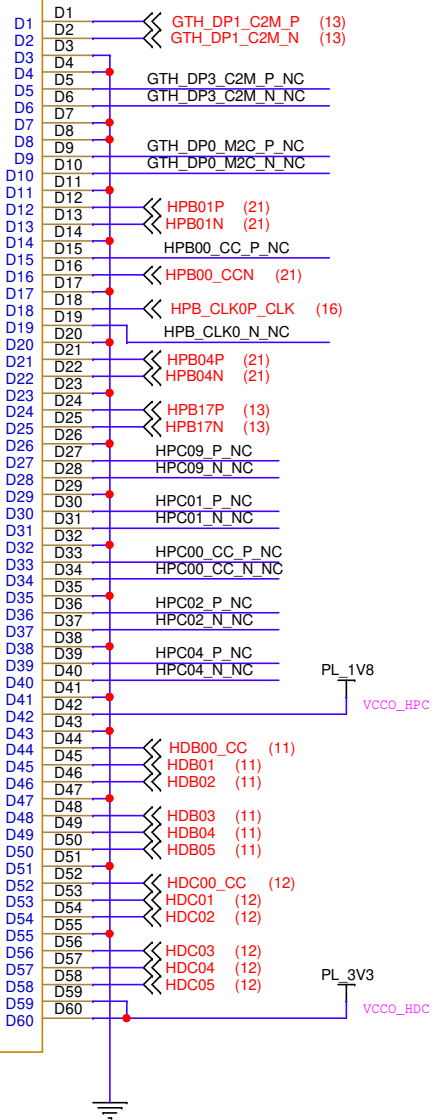
JA2B
ADM6-60-01.5-L-4-2-A-TR



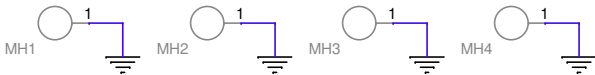
JA2C
ADM6-60-01.5-L-4-2-A-TR



JA2D
ADM6-60-01.5-L-4-2-A-TR

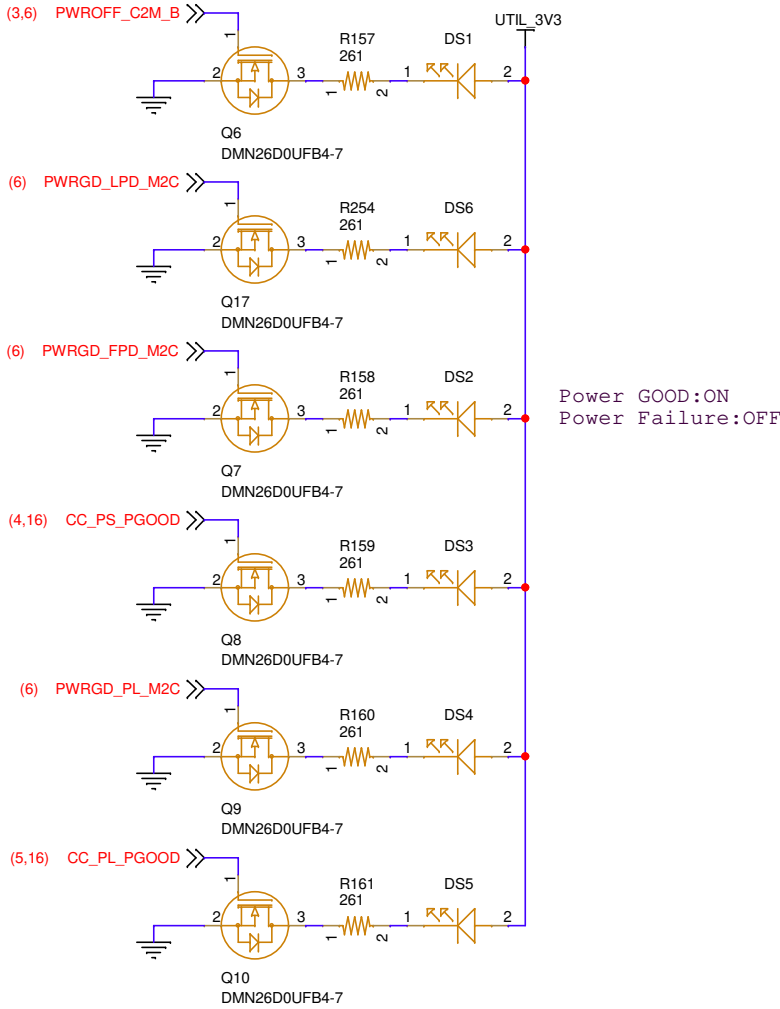


FOUR CORNER MOUNTING HOLES

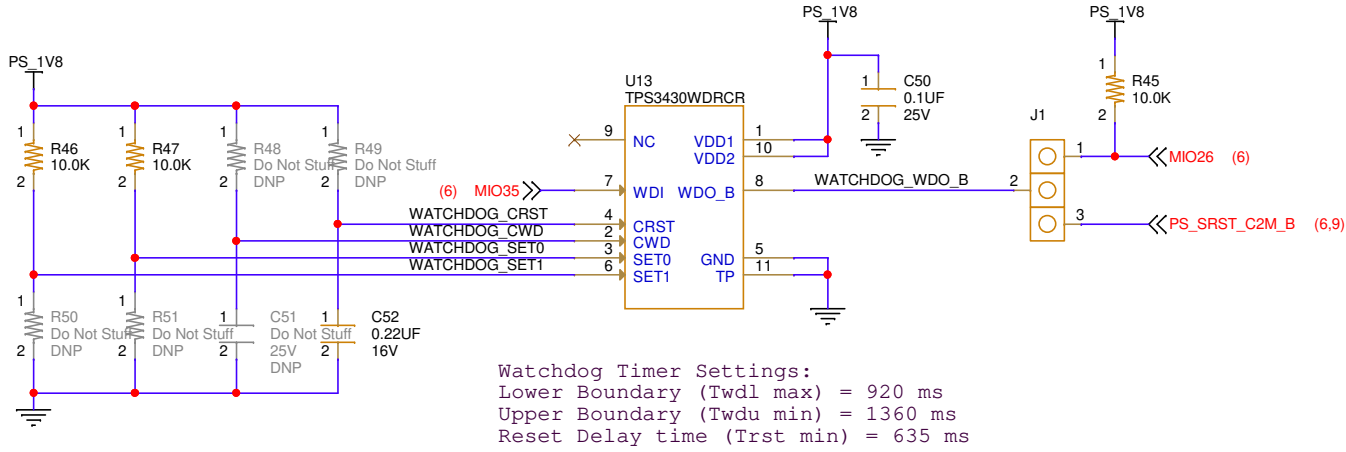


 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 07 - SOM240_2 Connector			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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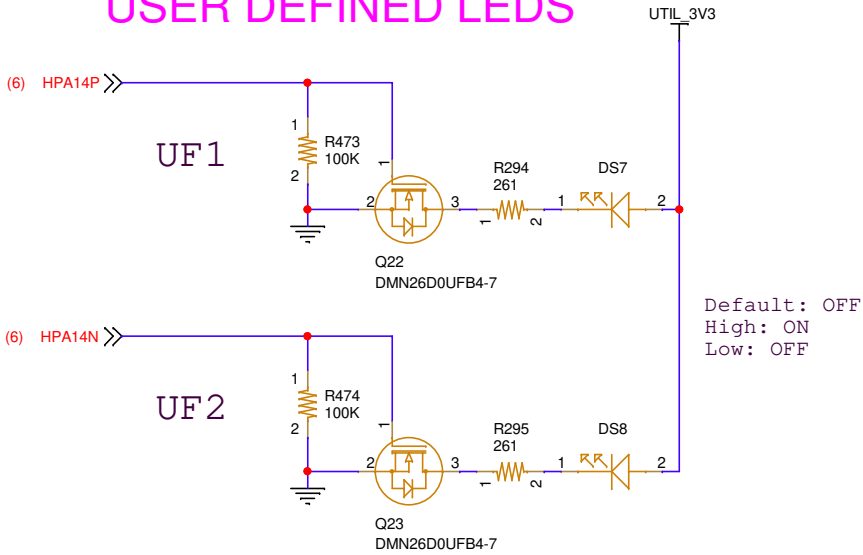
POWER STATUS LEDs



WATCHDOG TIMER

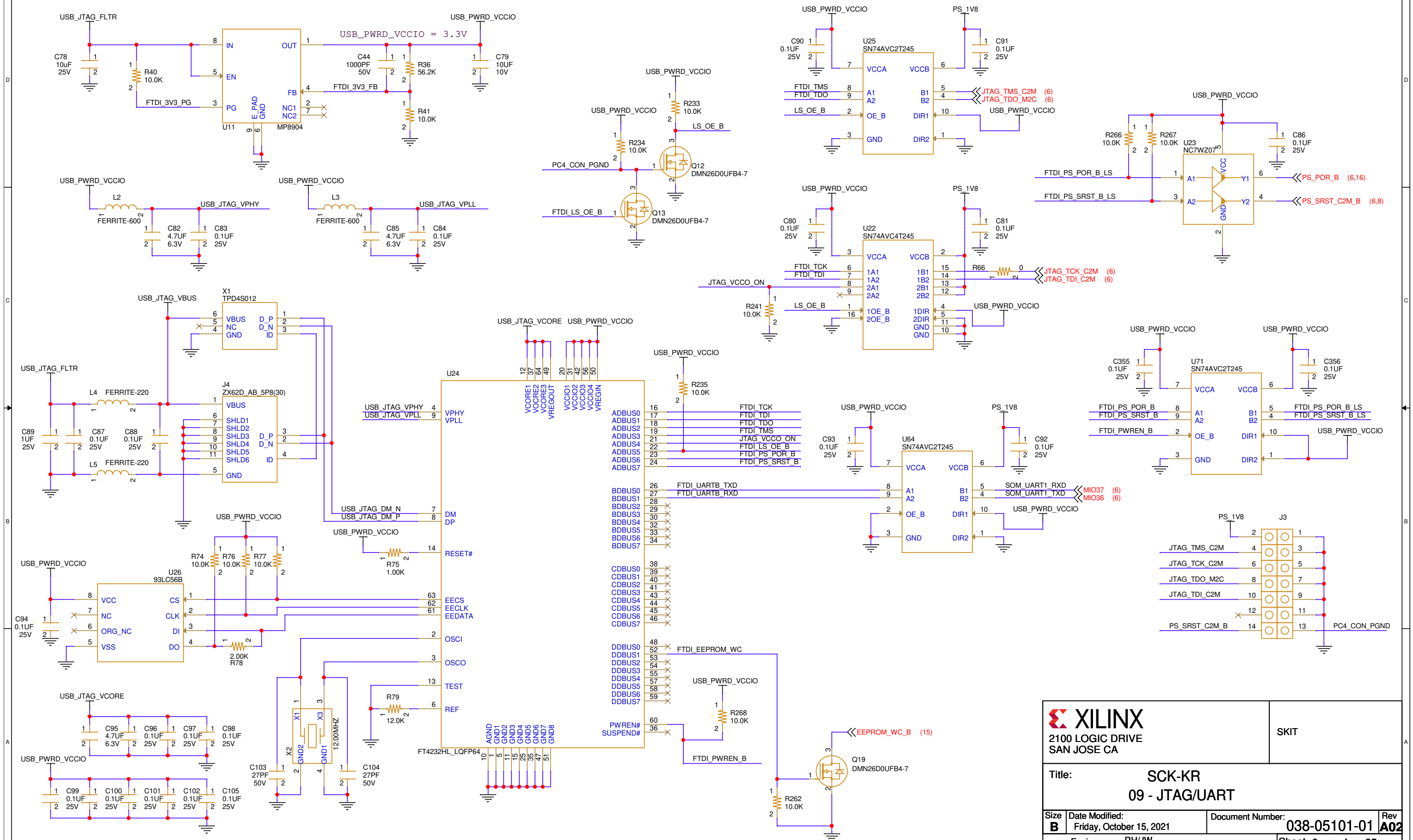


USER DEFINED LEDs



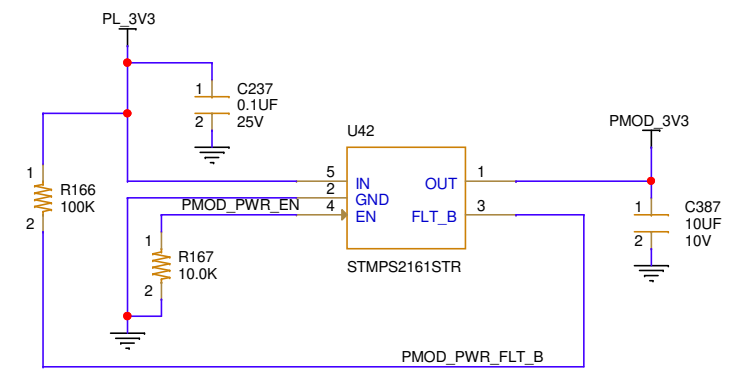
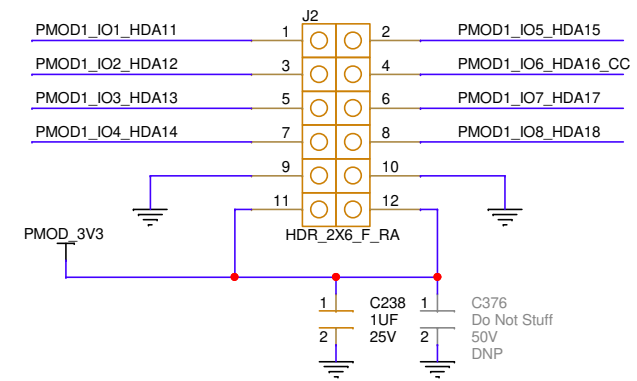
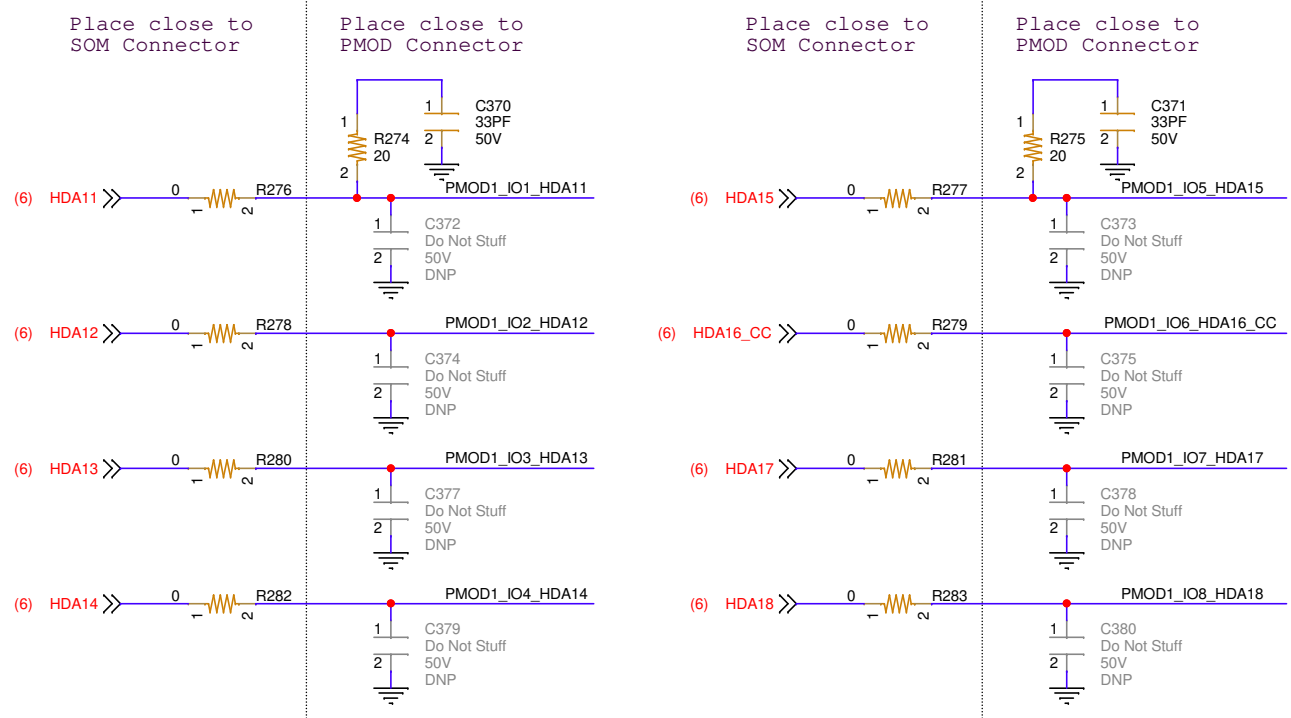
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Title: SCK-KR 08 - WD, LEDs			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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FTDI JTAG/UART

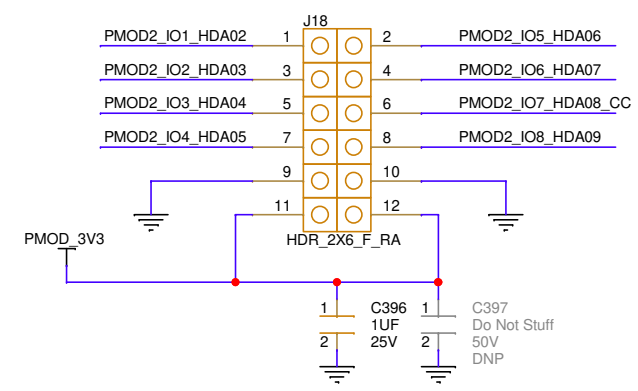
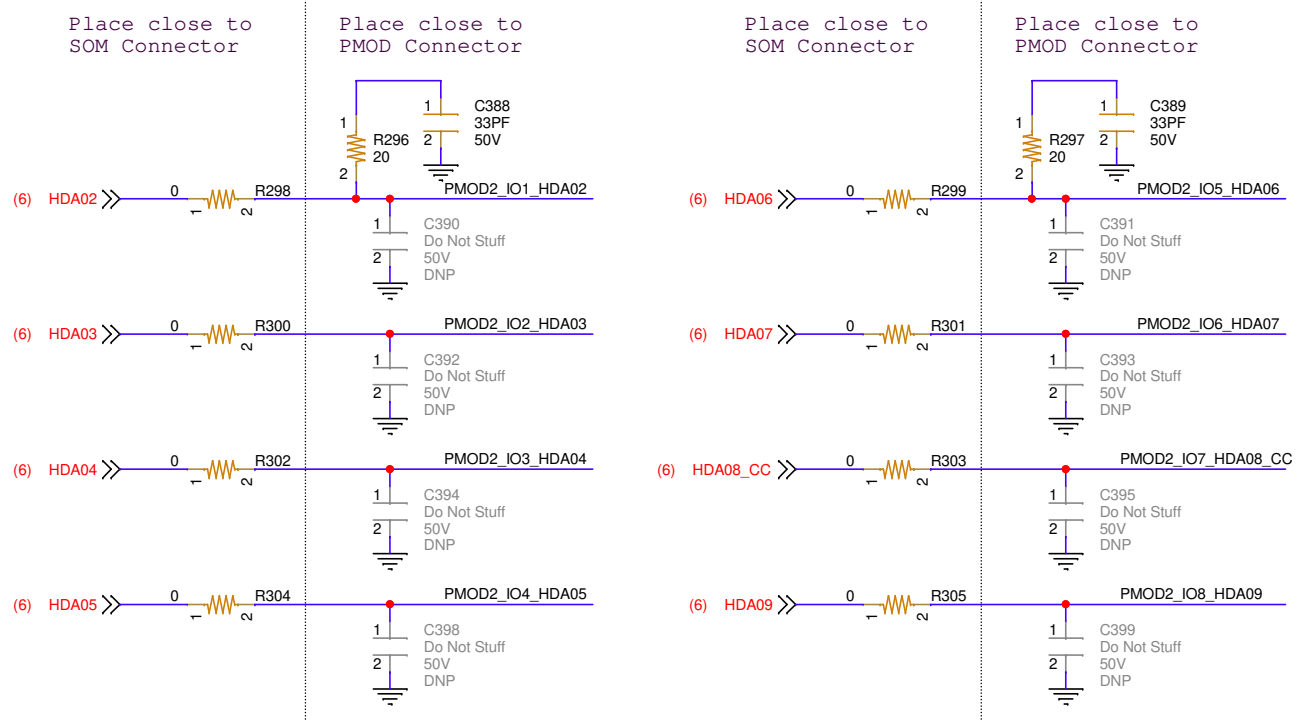


 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 09 - JTAG/UART			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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PMOD #1

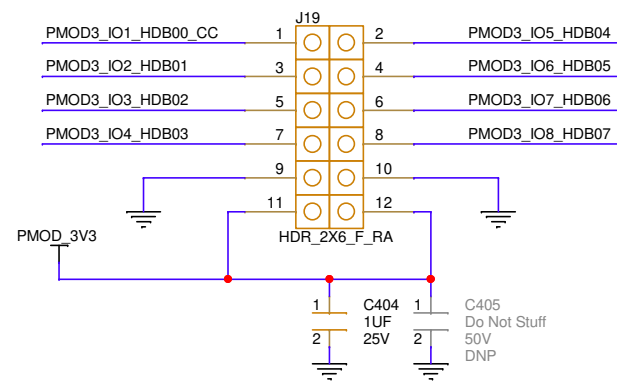
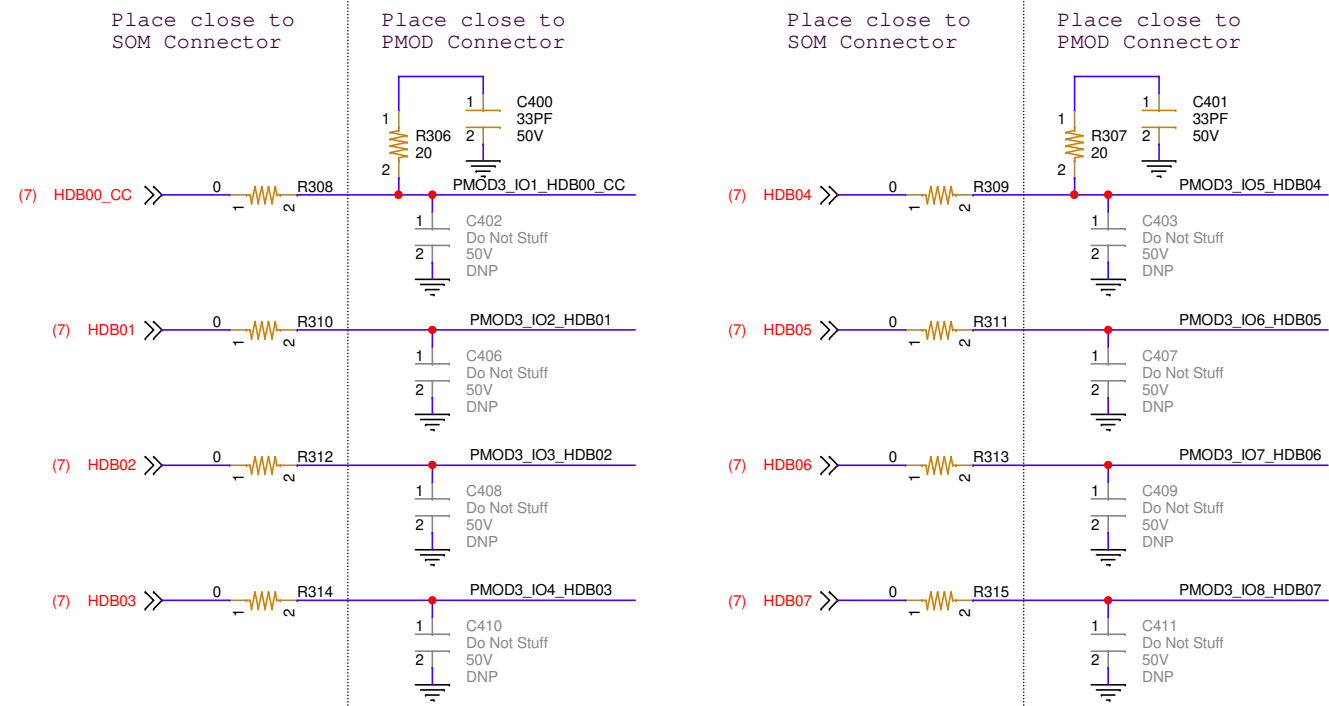


PMOD #2

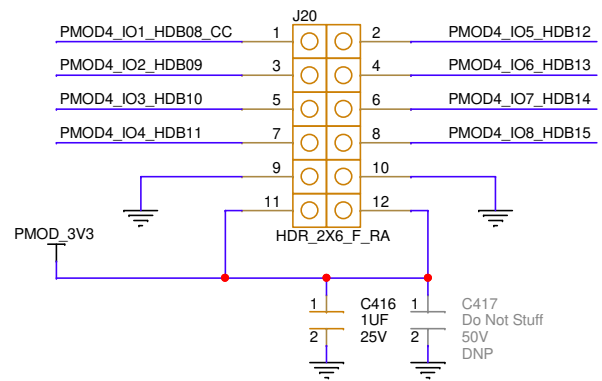
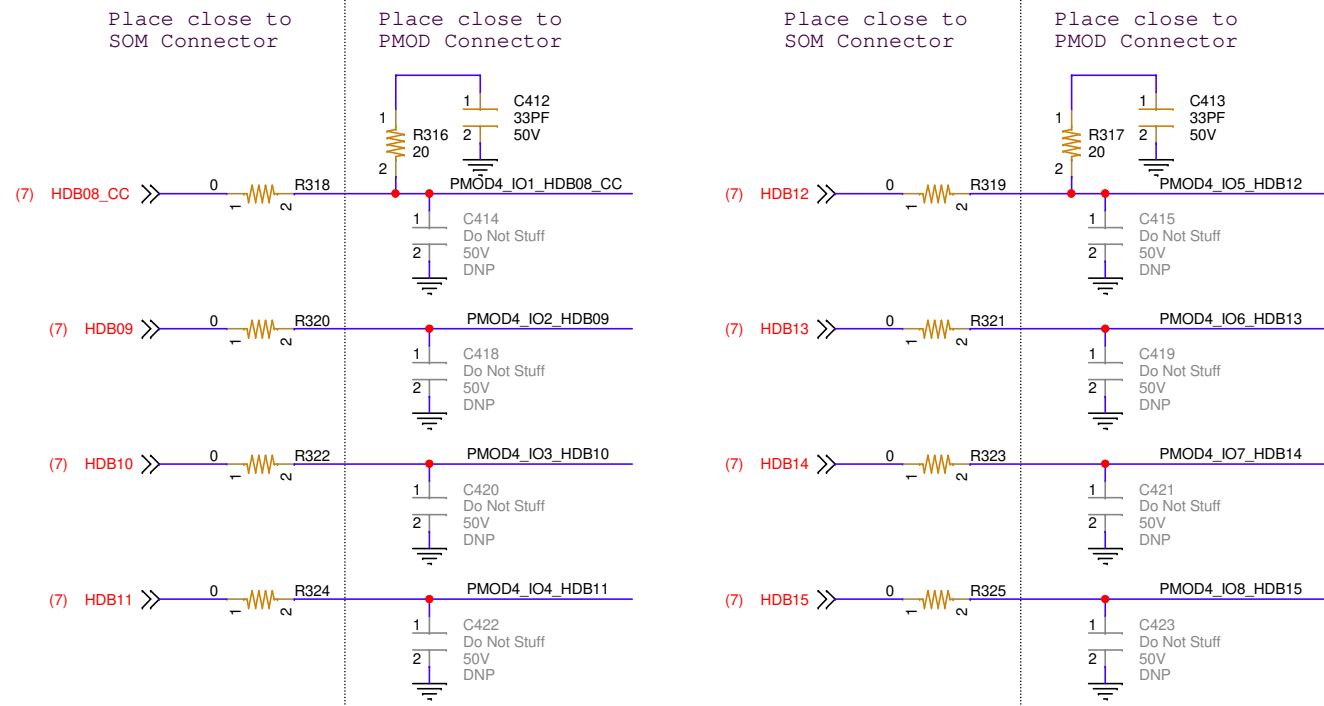


		SKIT	
Title: SCK-KR 10 - PMOD1&2			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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PMOD #3

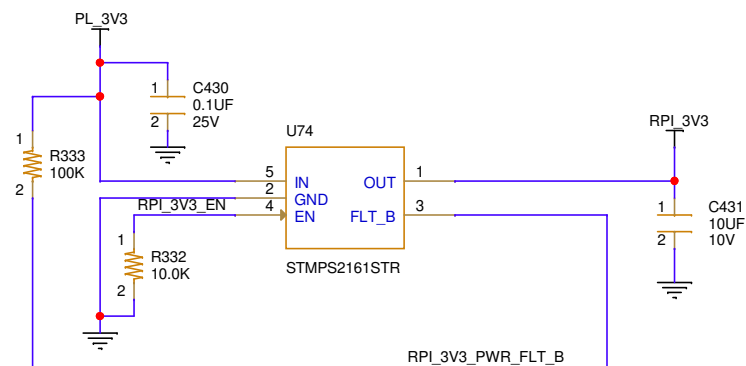
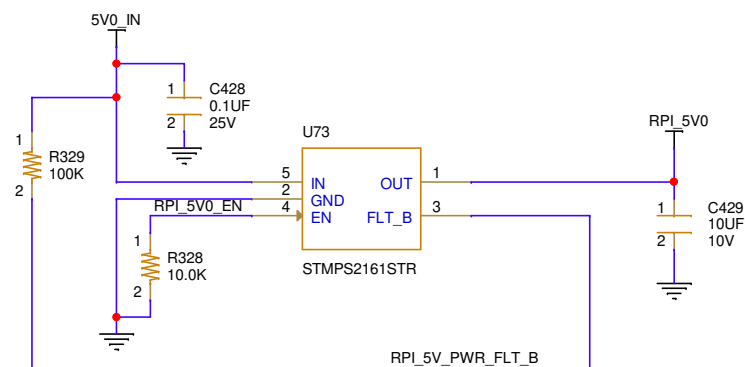
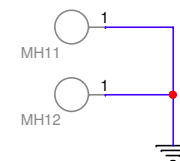
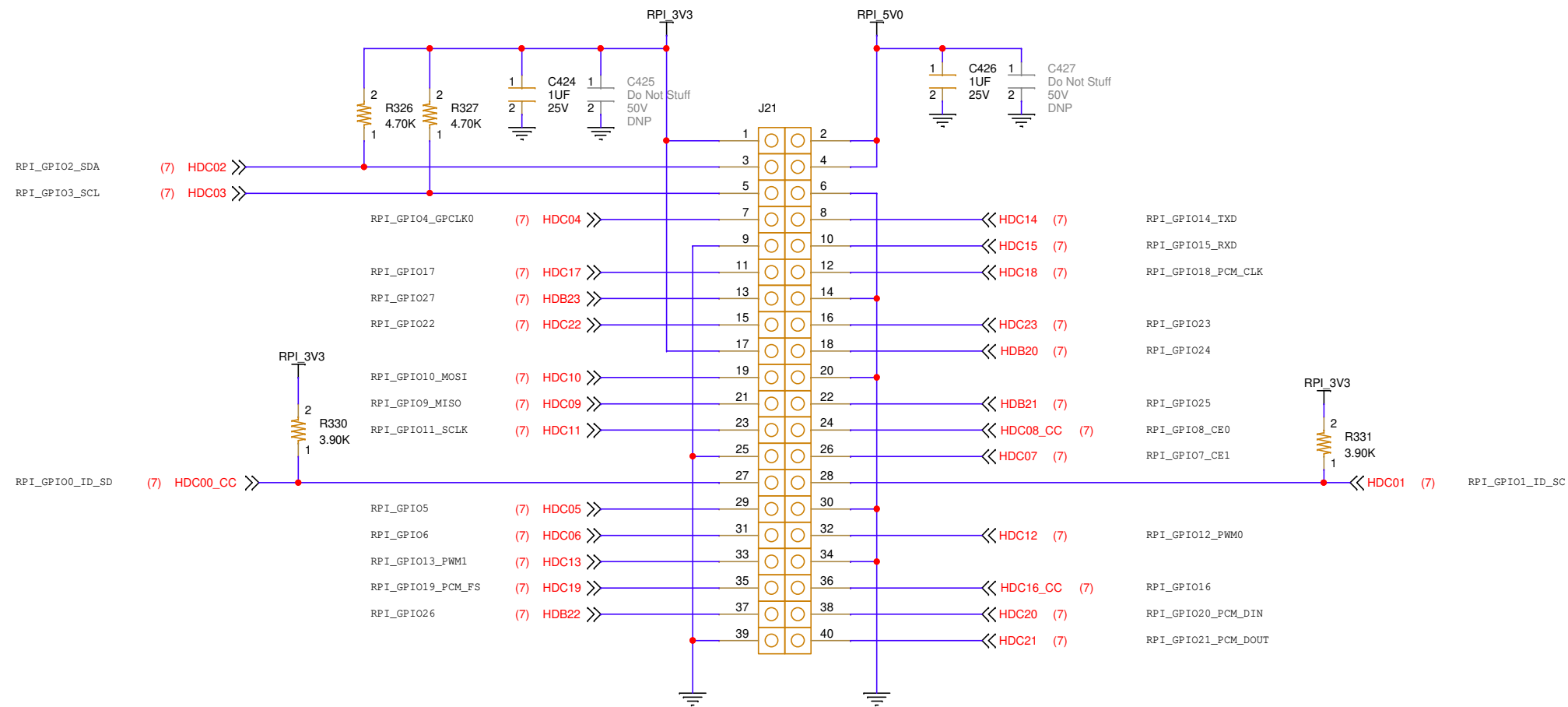


PMOD #4



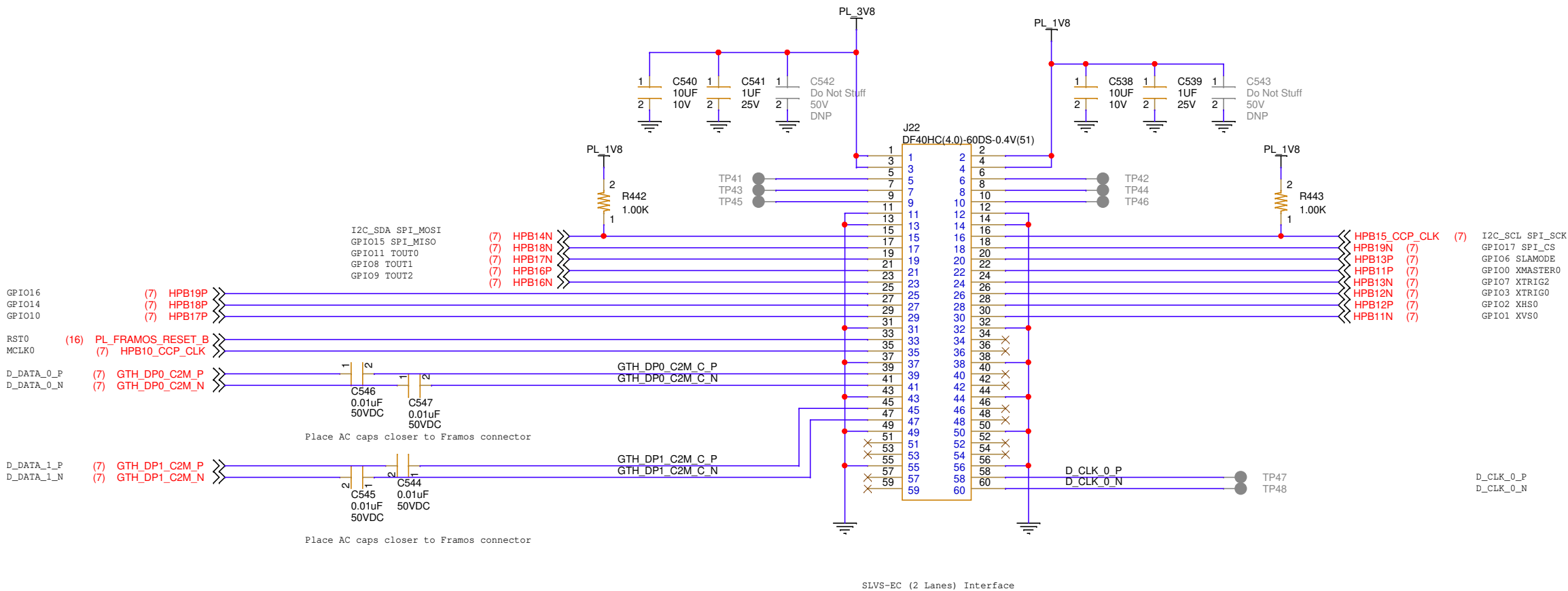
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Title: SCK-KR 11 - PMOD3&4			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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Raspberry Pi Expansion Header



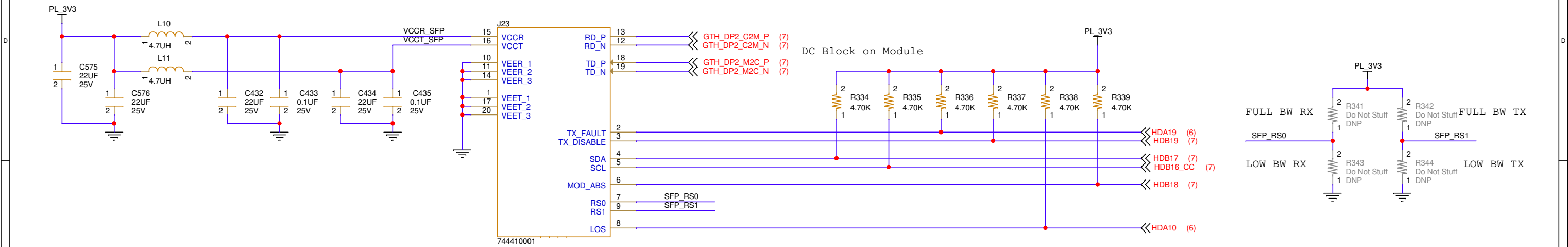
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Title: SCK-KR 12 - RPI Expansion CONN			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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Framos FPA SLVS-EC

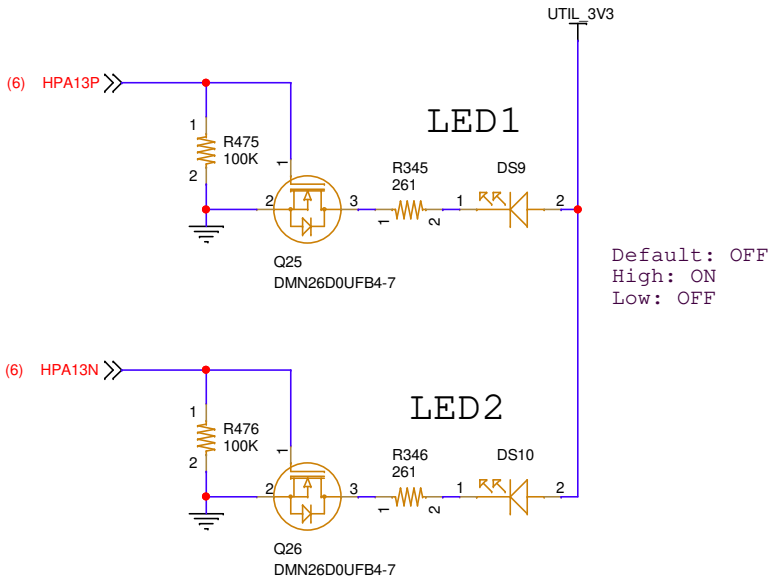


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Title: SCK-KR 13 - Framos CONN			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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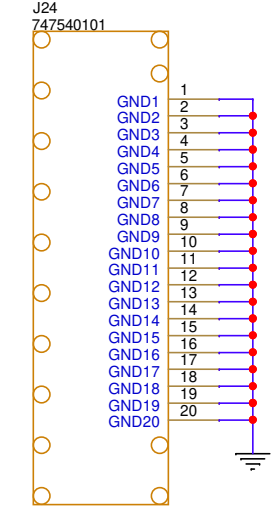
SFP+ Connector



SFP+ LEDs

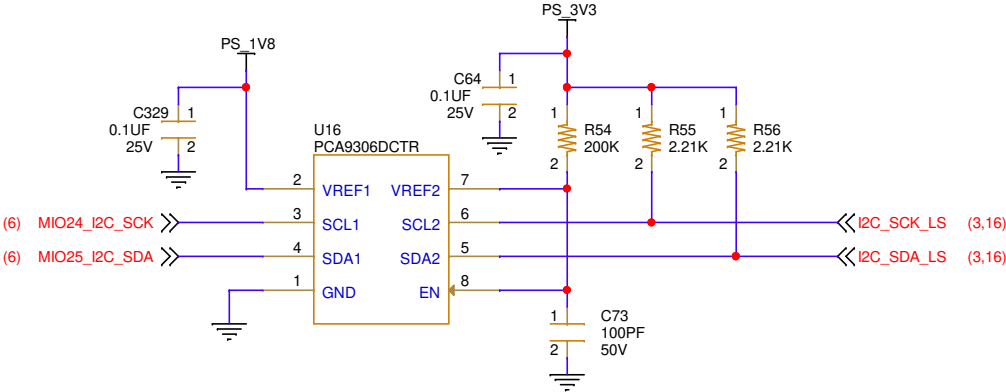


SFP+ Cage

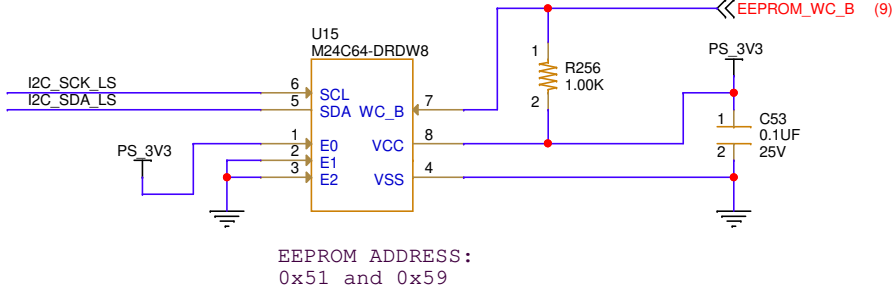


 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 14 - SFP+			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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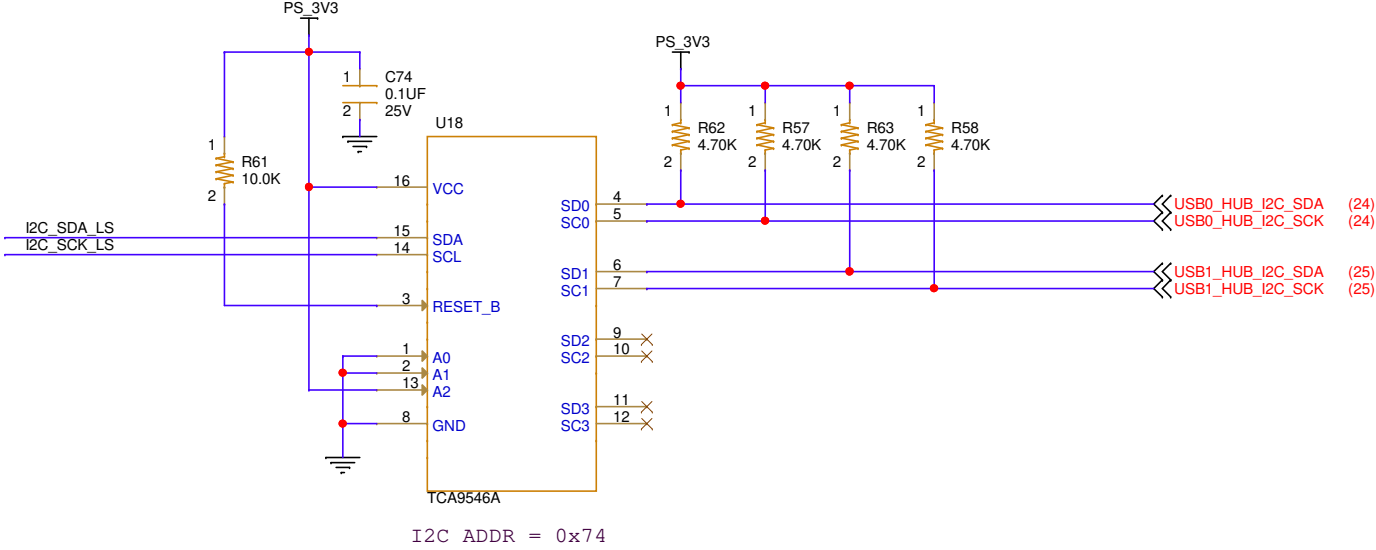
MIO I2C LEVEL SHIFTER



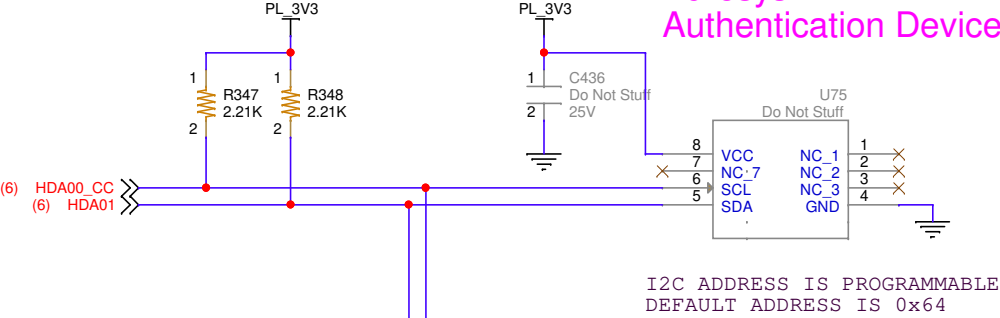
SYSTEM EEPROM



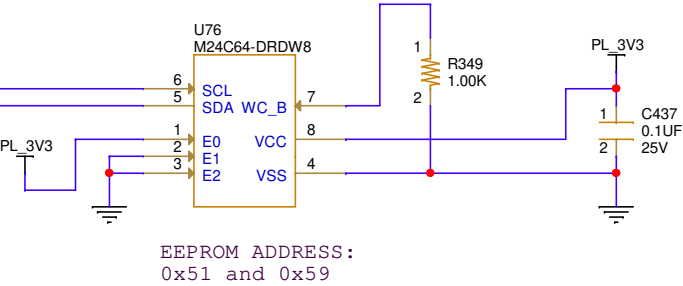
MIO I2C SWITCH



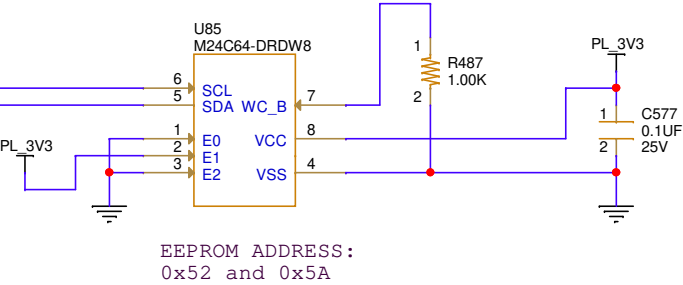
Euresys IP Authentication Device



Euresys IP EEPROM

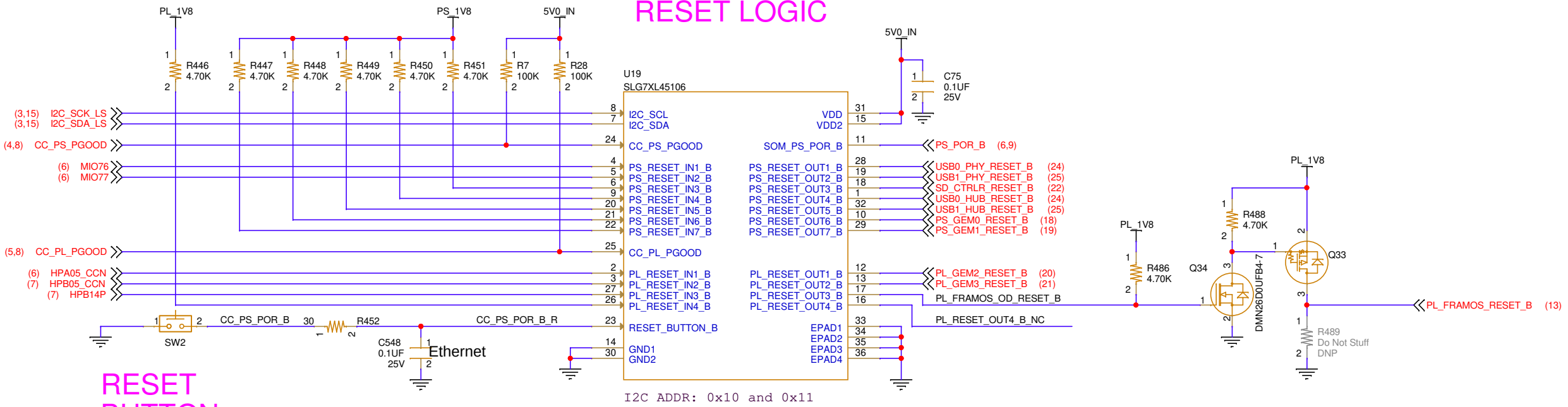


EtherCAT IP EEPROM



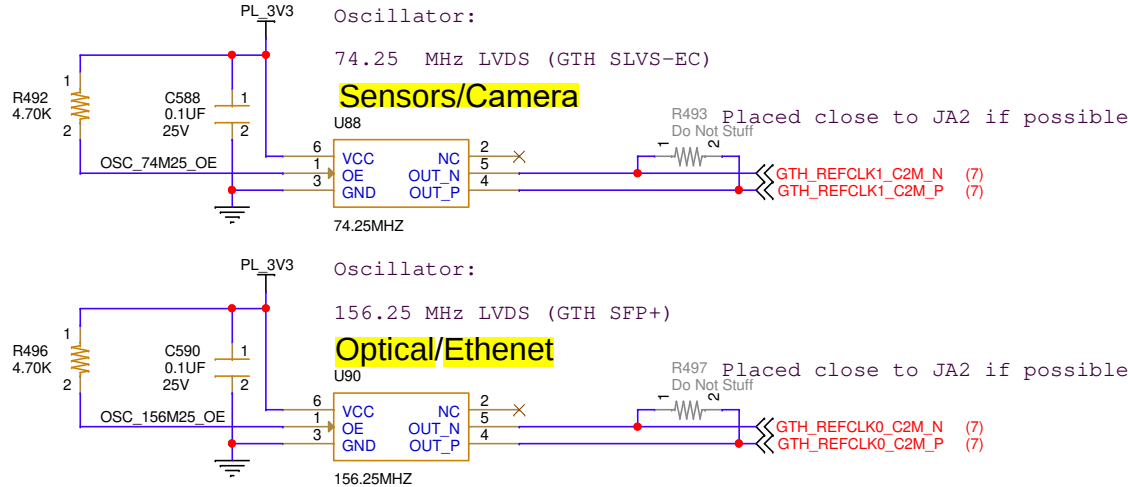
 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 15 - I2C, EEPROM			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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RESET LOGIC

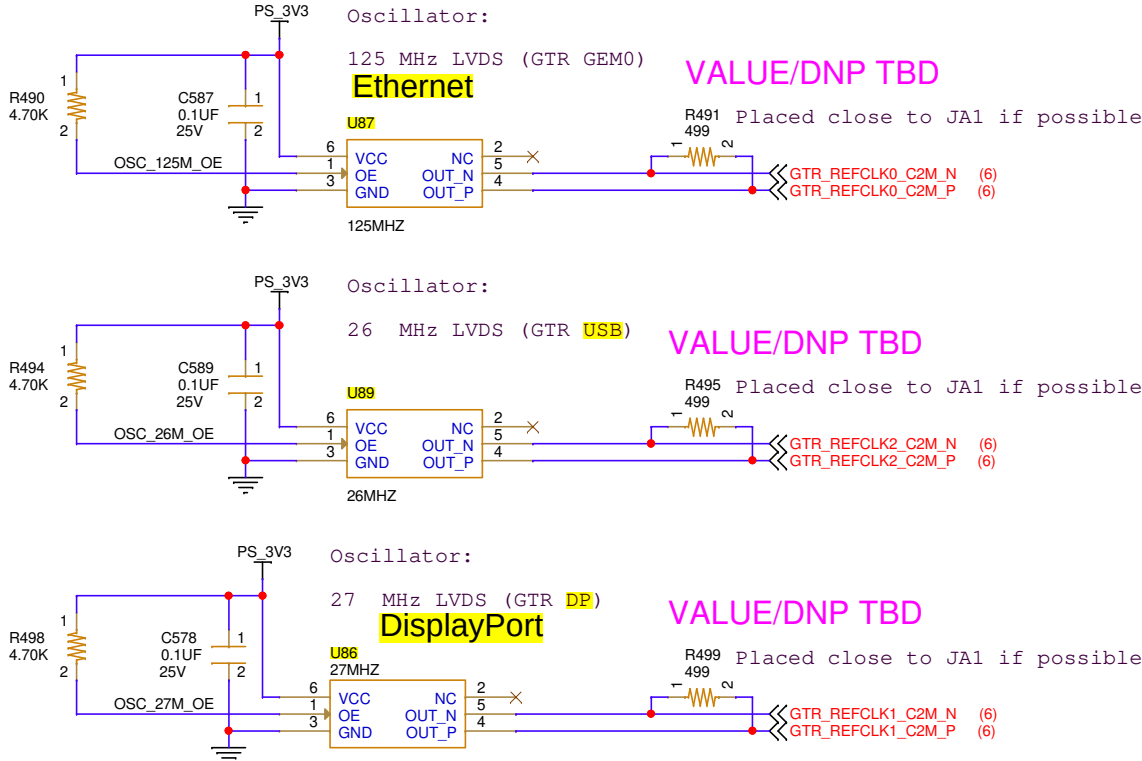


RESET BUTTON

PL



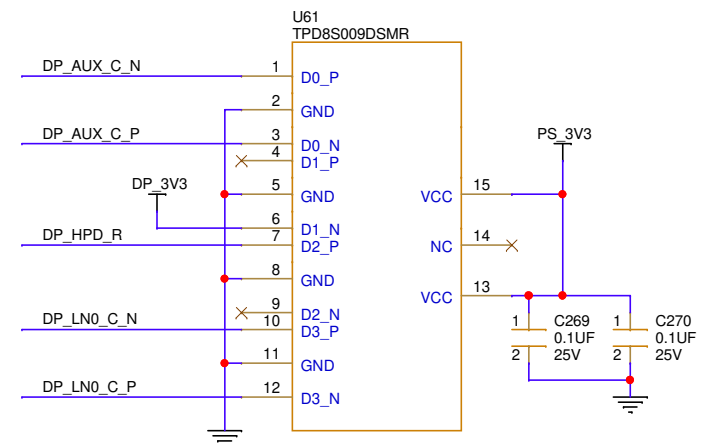
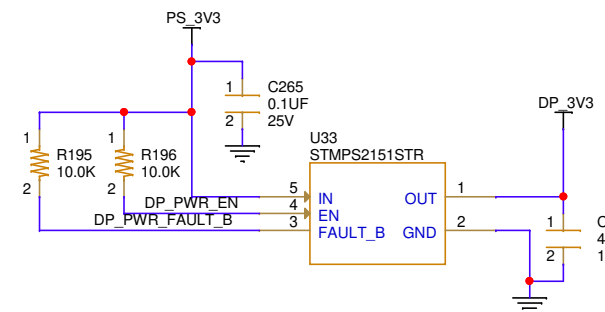
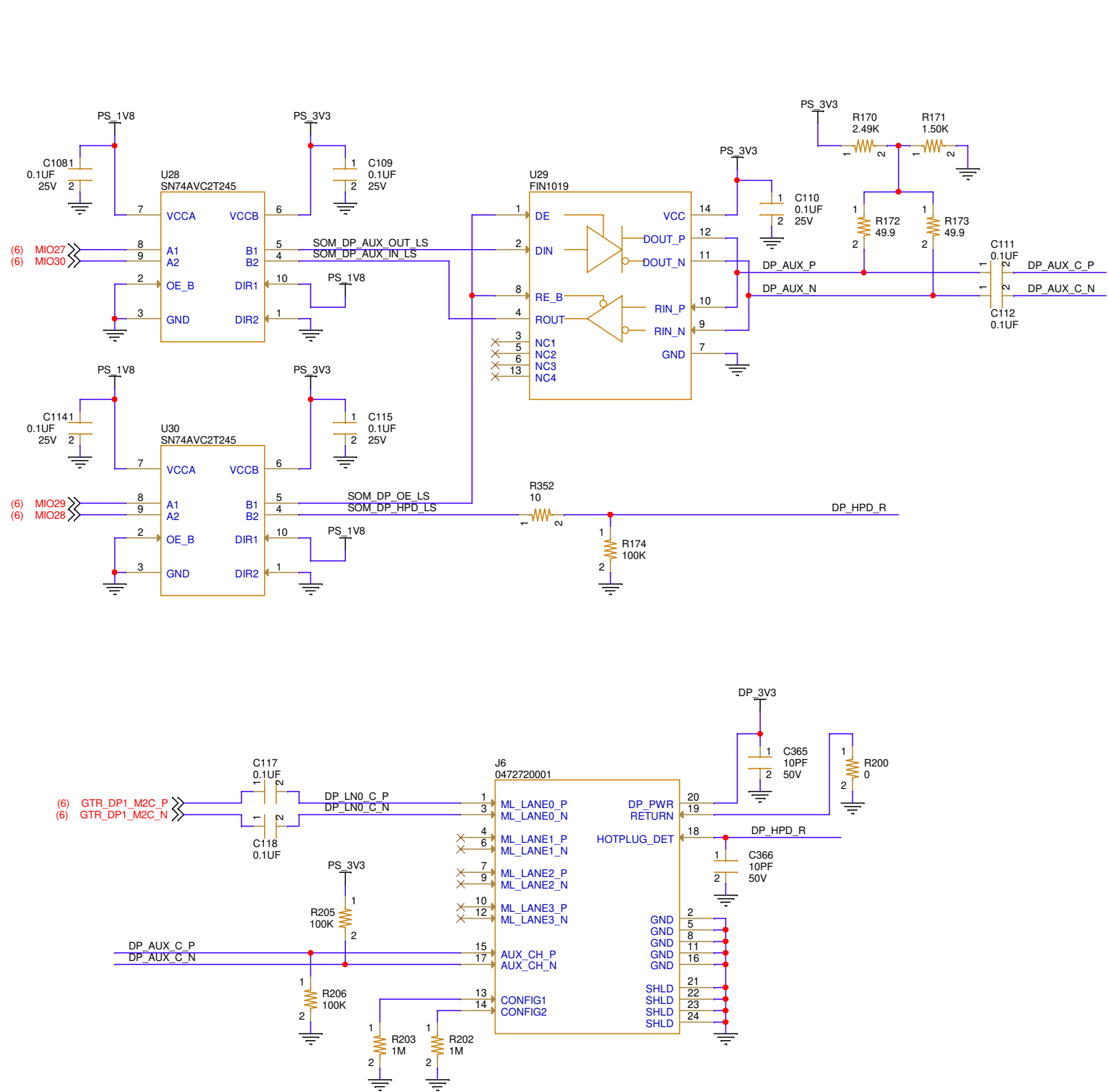
PS



 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 16_Clock Gen, Reset			
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High Performance Bank A
Gigabit Ethernet PHY 2
High Performance Bank B
Gigabit Ethernet PHY 3

DISPLAY PORT

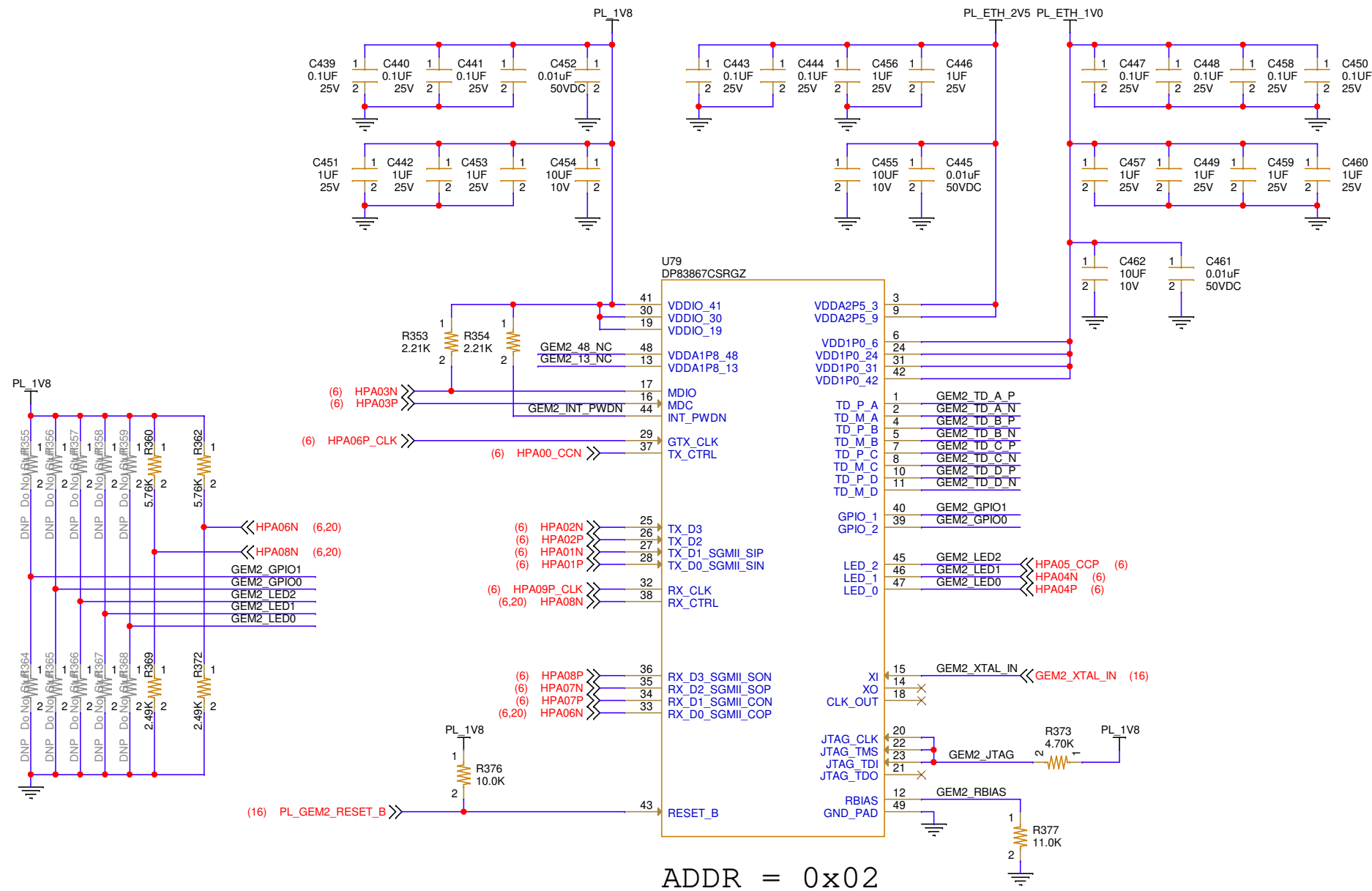


 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 17_DisplayPort			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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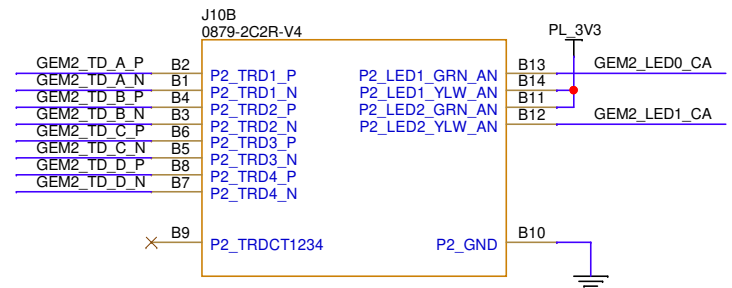
A vertical line with points A, B, C, and D marked from bottom to top. A horizontal arrow points to point C.



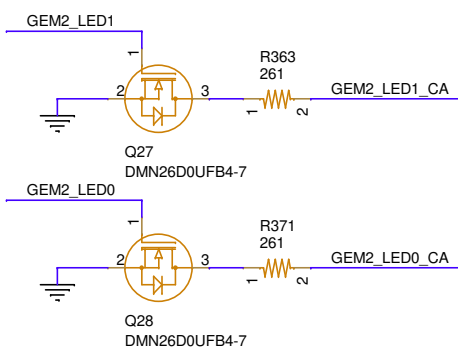
PL GEM2 RGMII ETHERNET



ADDR = 0x02

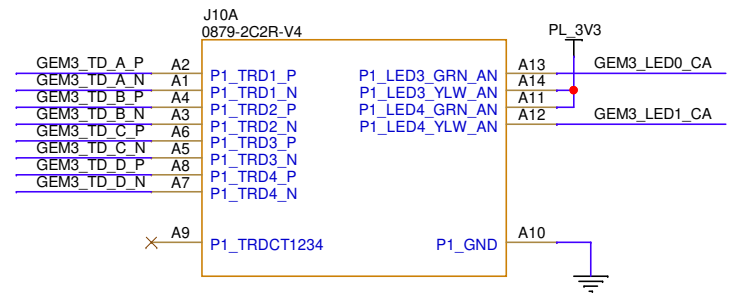
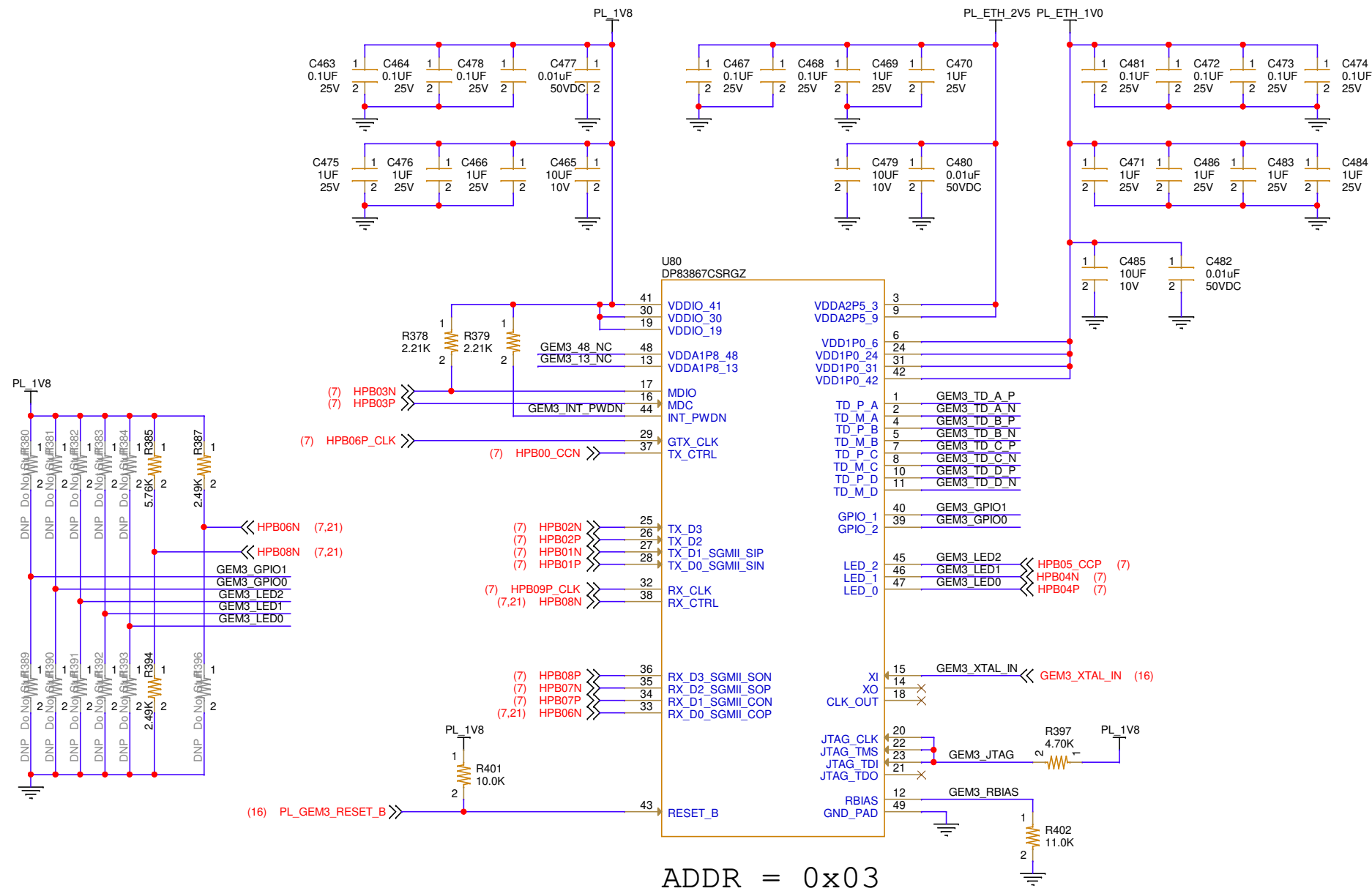


TOP LEFT

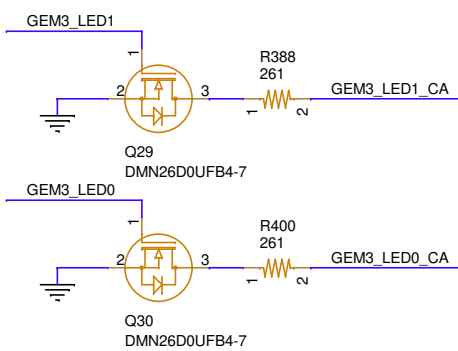


 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 20_PL_GEM2			
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PL GEM3 RGMII ETHERNET

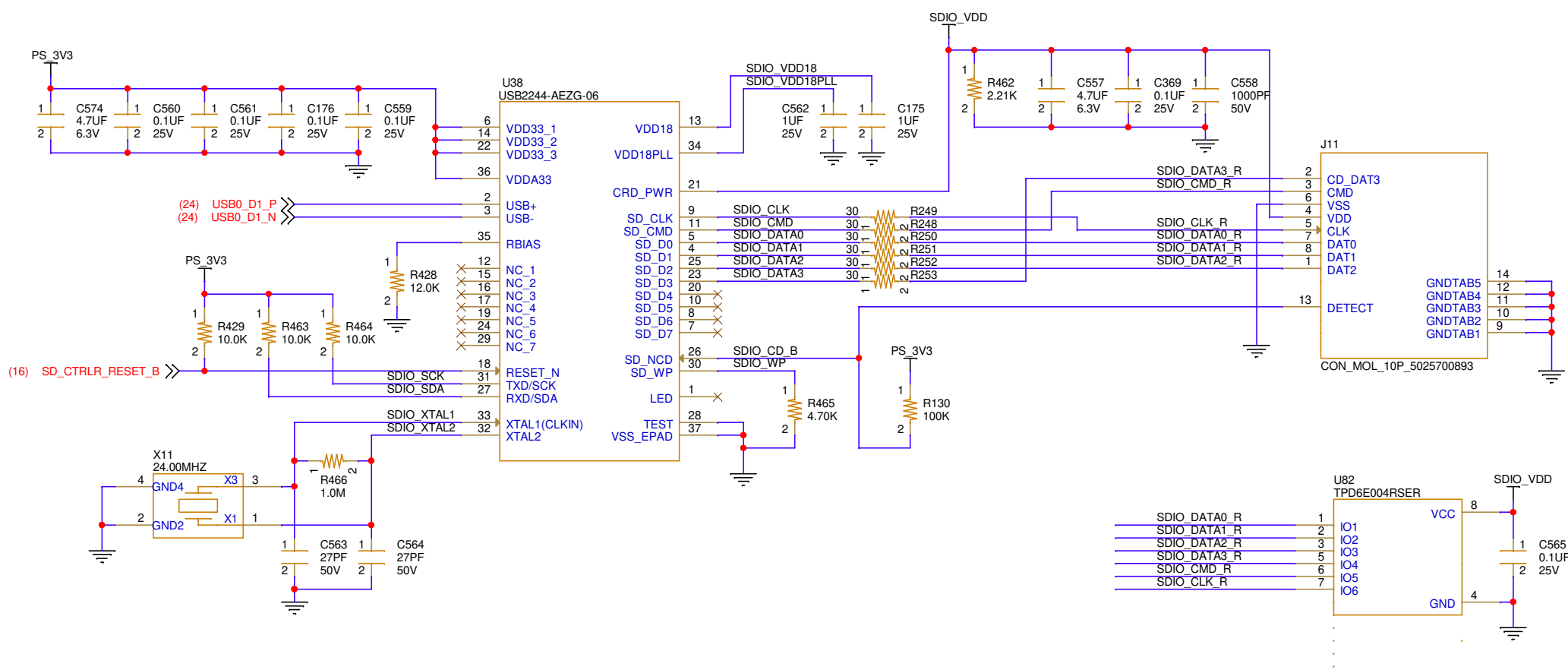


BOTTOM LEFT

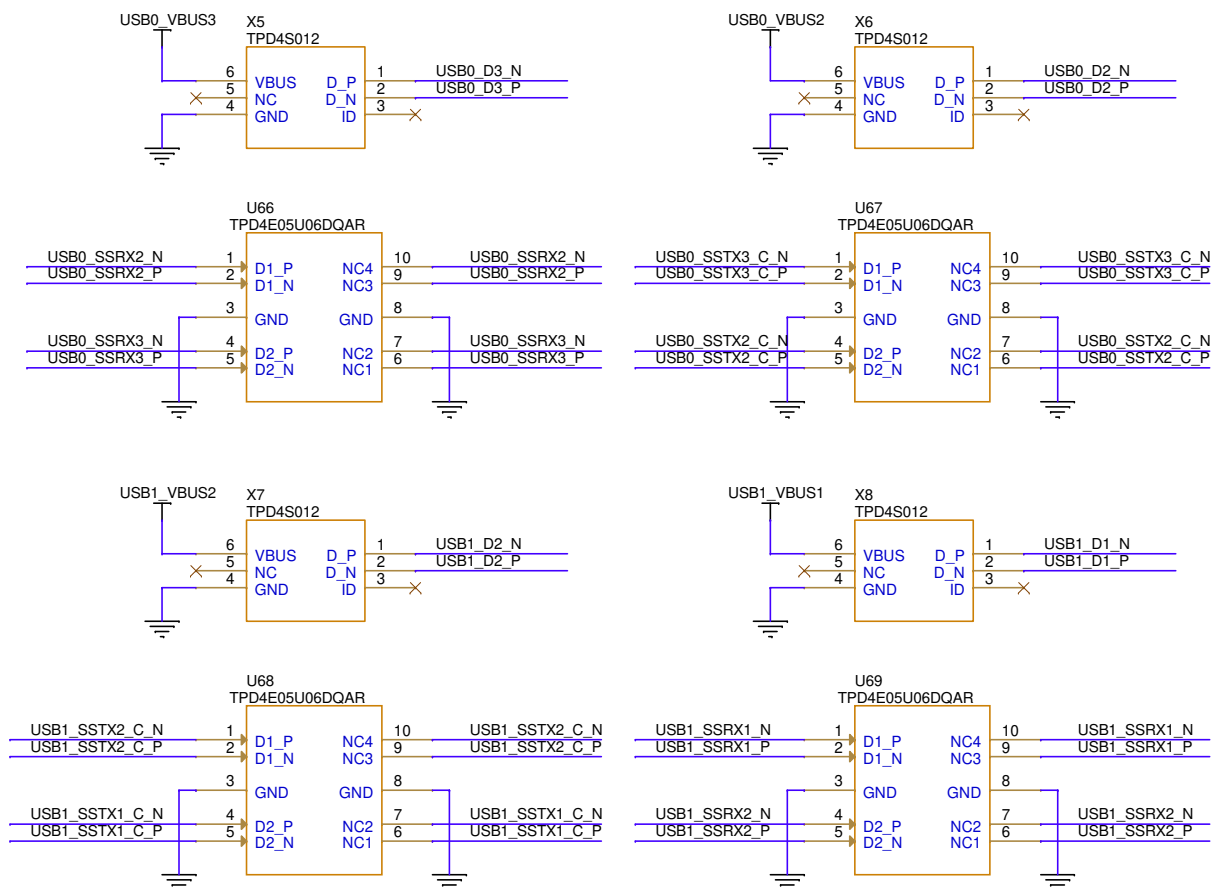
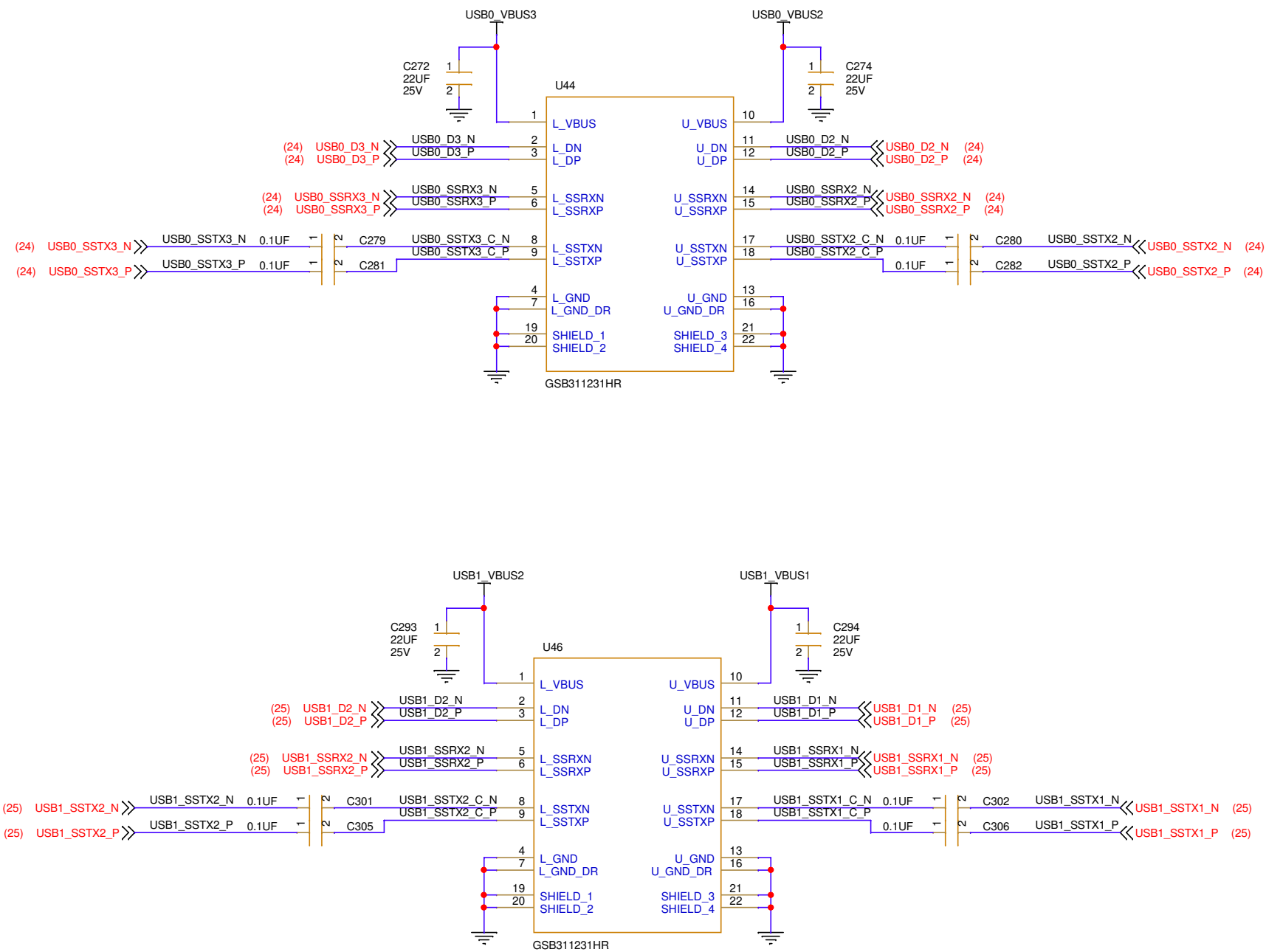


 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 21_PL_GEM3			
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SD 2.0

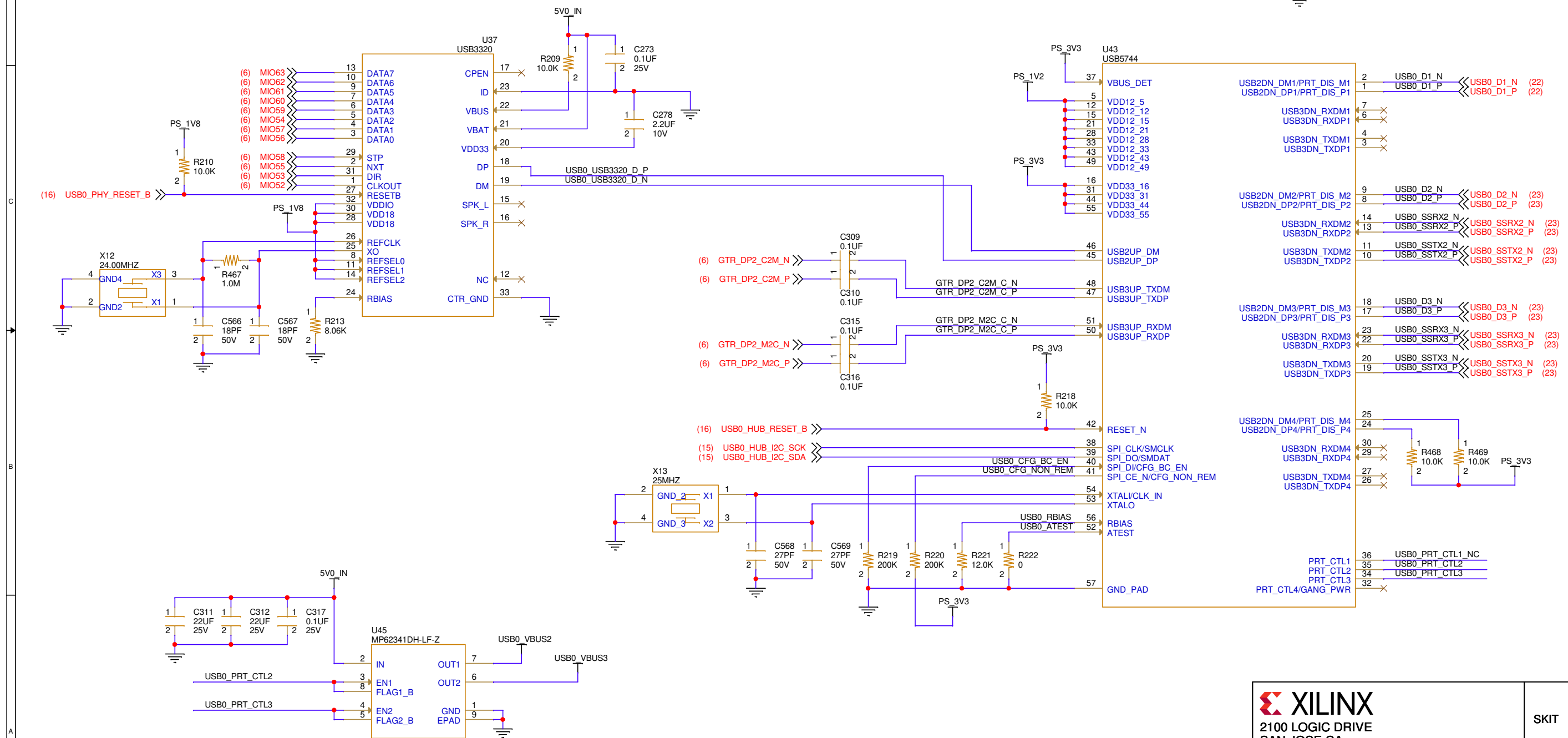


USB CONNECTORS



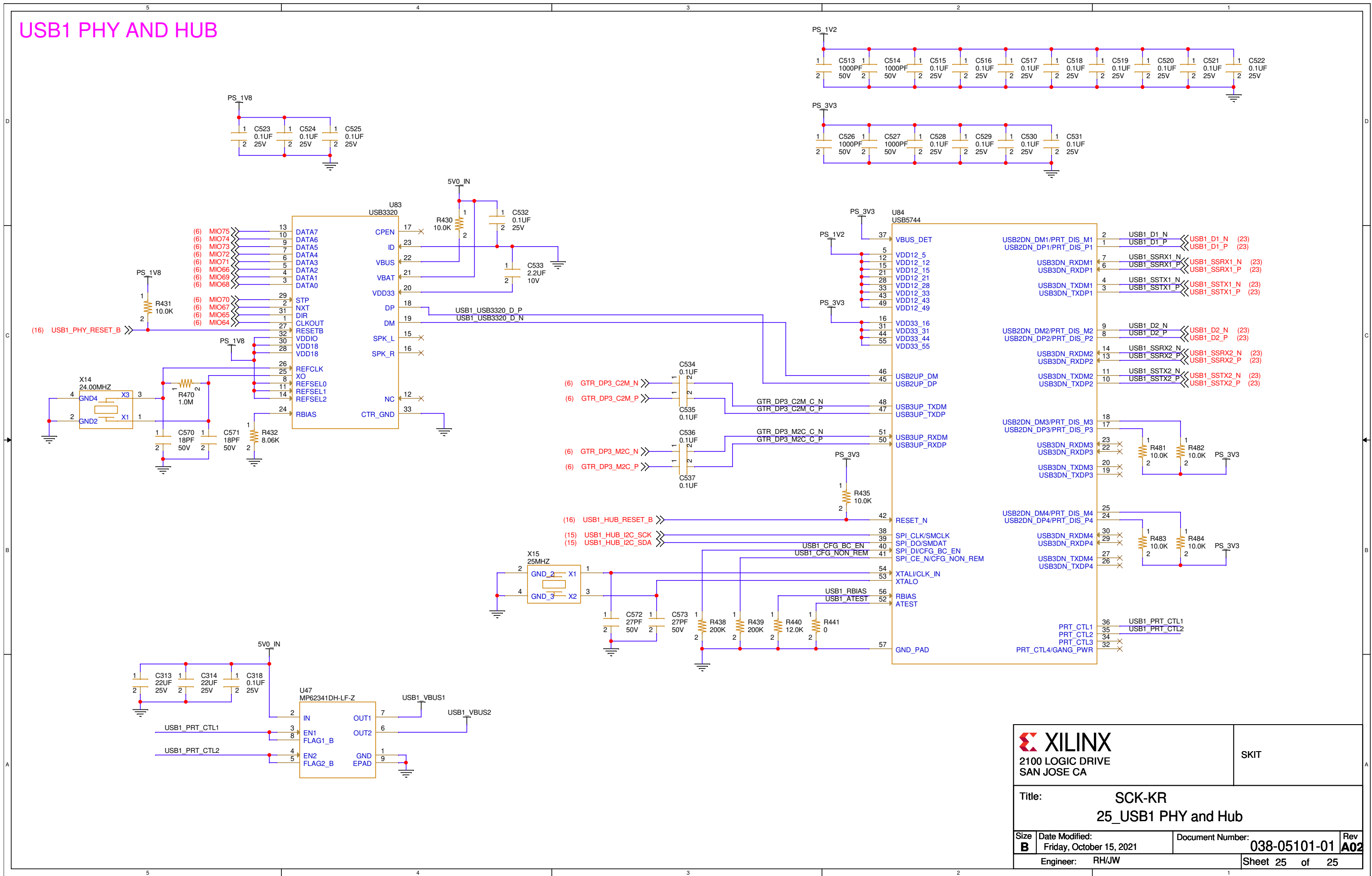
 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 23_ USB Connectors			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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USB0 PHY AND HUB



 XILINX 2100 LOGIC DRIVE SAN JOSE CA		SKIT
Title: SCK-KR 24_USB0 PHY and Hub		
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01
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USB1 PHY AND HUB



 2100 LOGIC DRIVE SAN JOSE CA		SKIT	
Title: SCK-KR 25_USB1 PHY and Hub			
Size B	Date Modified: Friday, October 15, 2021	Document Number: 038-05101-01	Rev A02
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